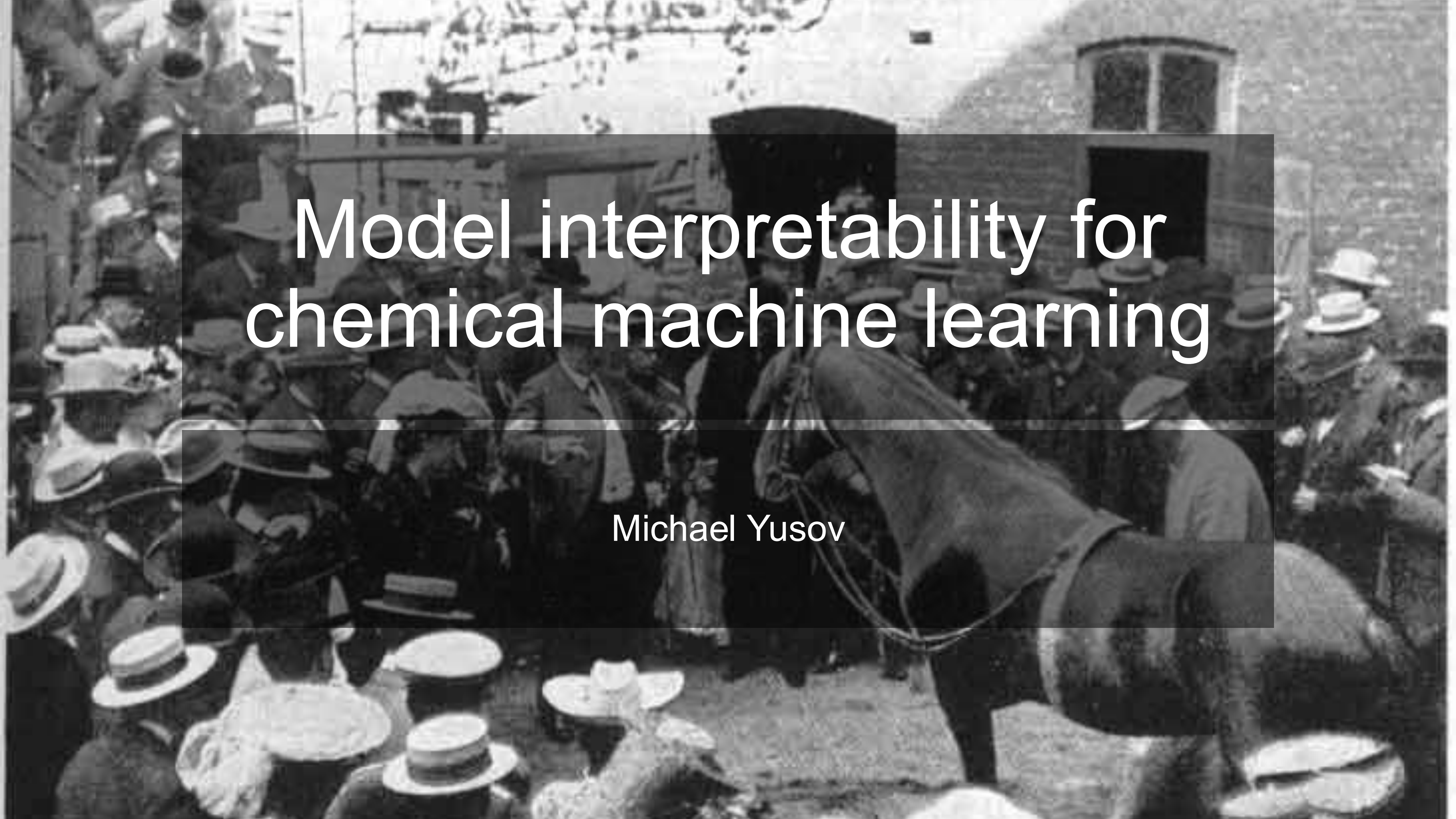


Model interpretability for chemical machine learning

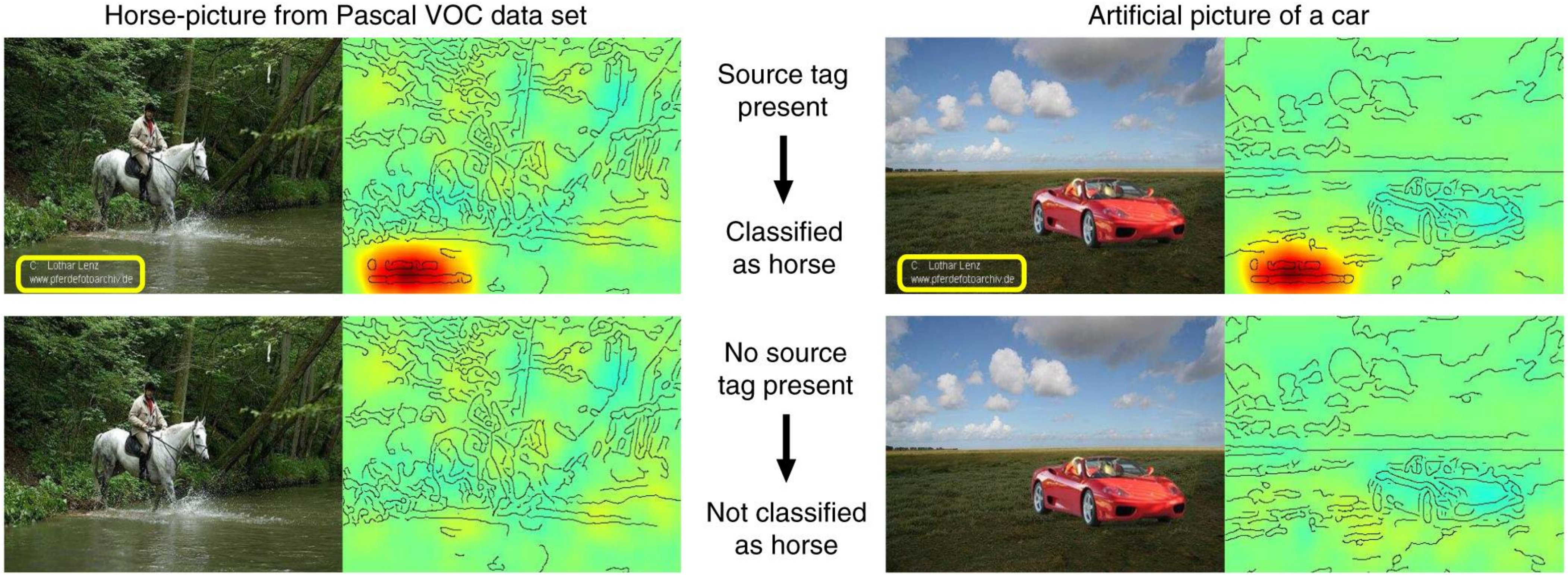
Michael Yusov

Clver Hans

“In the field of artificial intelligence, **the Clever Hans effect** describes a phenomenon where an algorithm seems to make correct predictions without having the relevant data and/or by using incorrect reasoning.” —Wikipedia

https://en.wikipedia.org/wiki/Clever_Hans

The Clever Hans effect in AI



Lapuschkin, S. *et al.* Unmasking Clever Hans predictors and assessing what machines really learn. *Nat. Commun.* **10**, 1096 (2019). https://doi.org/10.1038/s41467-019-08987-4

What is model interpretability?

Definition: **Model interpretability is the extent to which a human can understand how a machine learning model uses input features to produce predictions, including why a particular prediction was made and what relationships the model has learned.**

In this lecture, we use “interpretability” and “explainability” interchangeably.

Why study model interpretability for chemical machine learning?

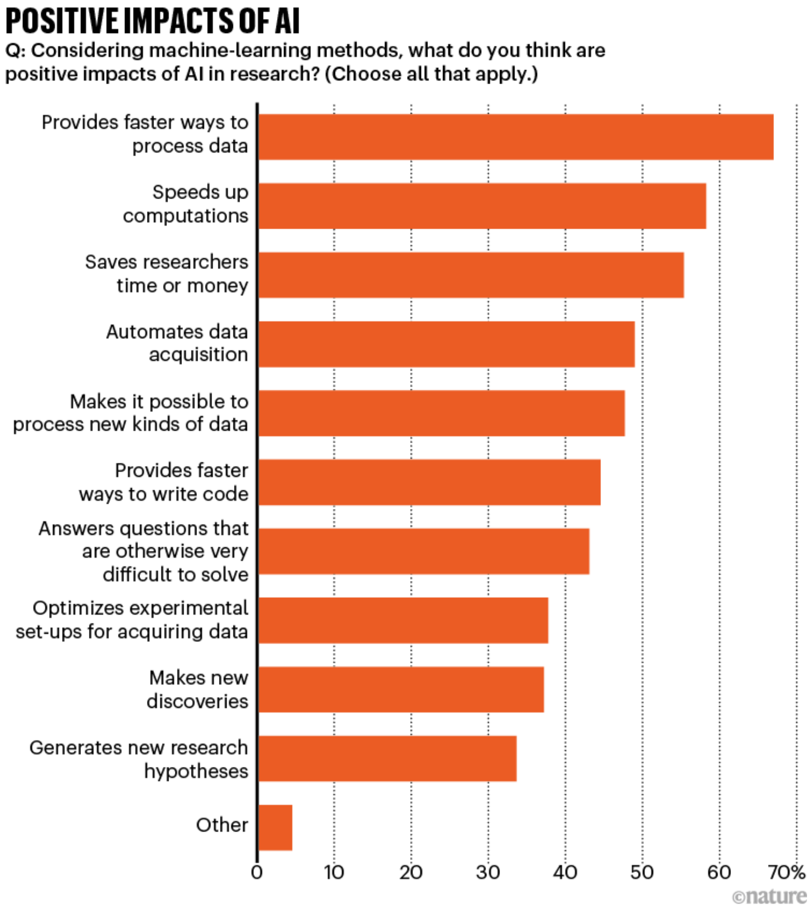
We study interpretability to:

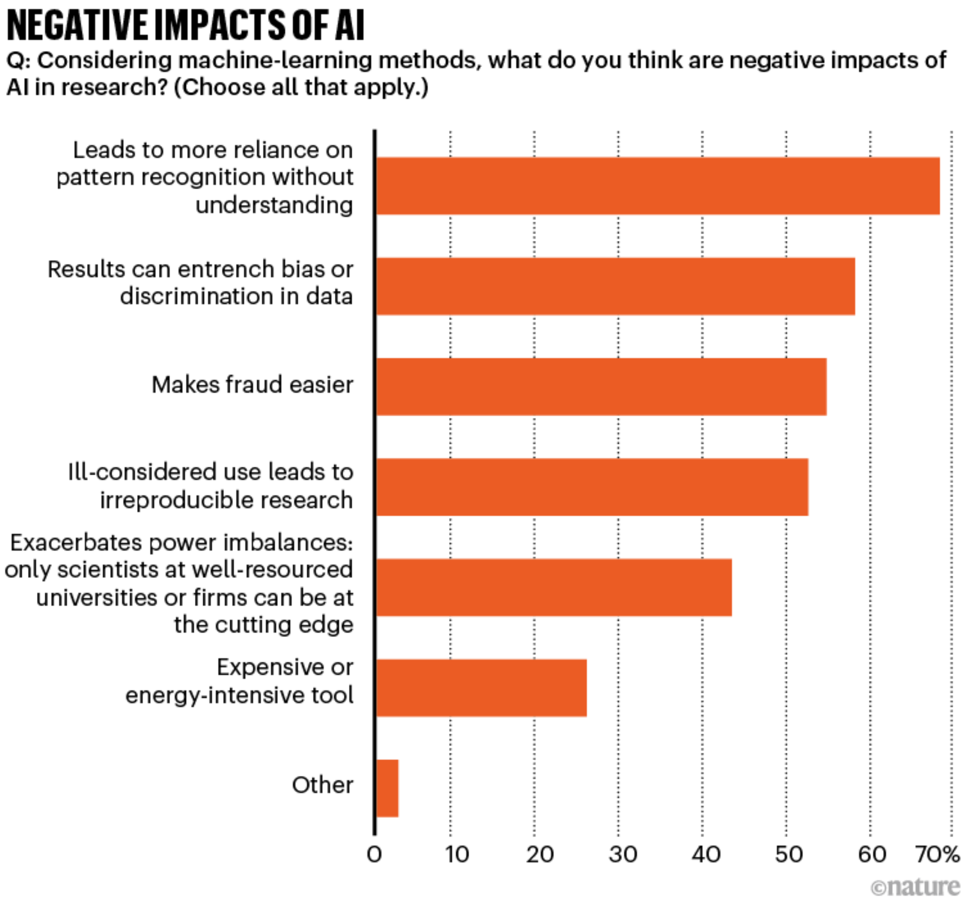
1. avoid blindly trusting predictive performance,
2. check whether the model learned meaningful features,
3. generate scientific insight,
4. make predictions actionable,
5. build trust with collaborators,
6. or debug models.

The goal of science is to gain knowledge; knowing why a model made certain predictions can help us learn more about the chemistry we are studying.

Interpretable machine learning allows the model itself to become the source of knowledge instead of the data.

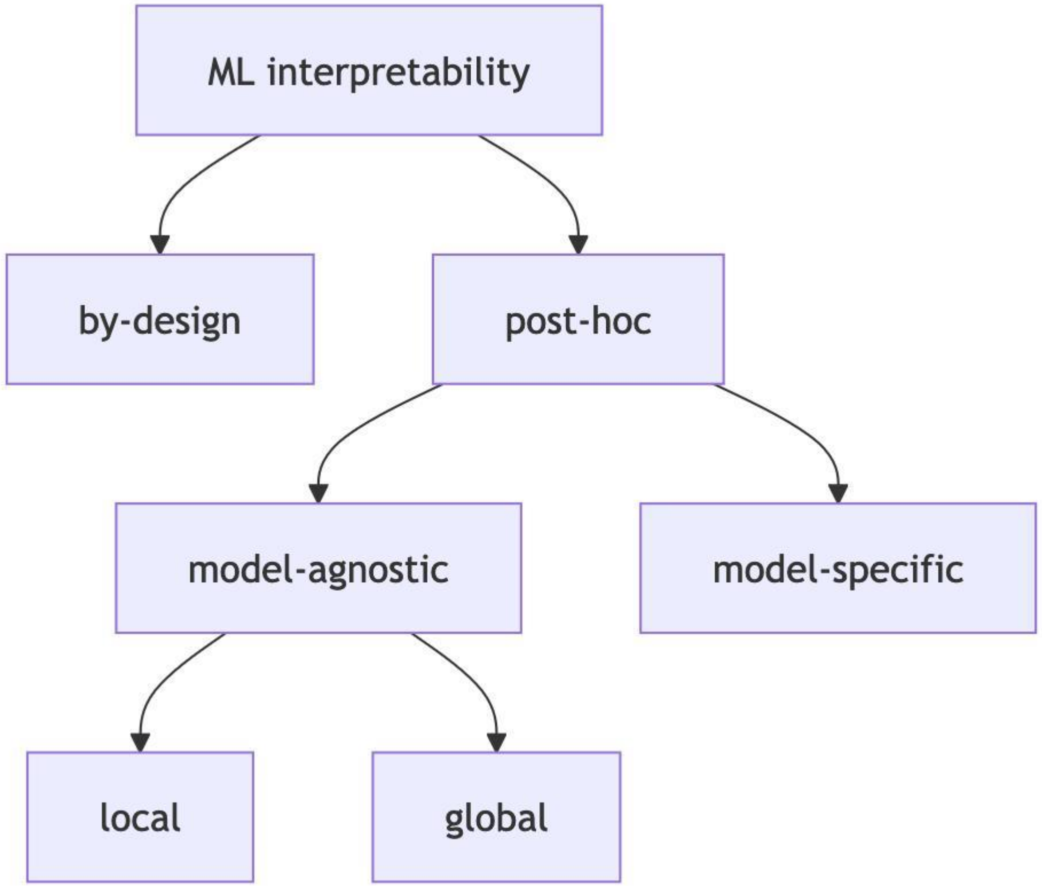
“AI and science: what 1,600 researchers think” (September 2023)





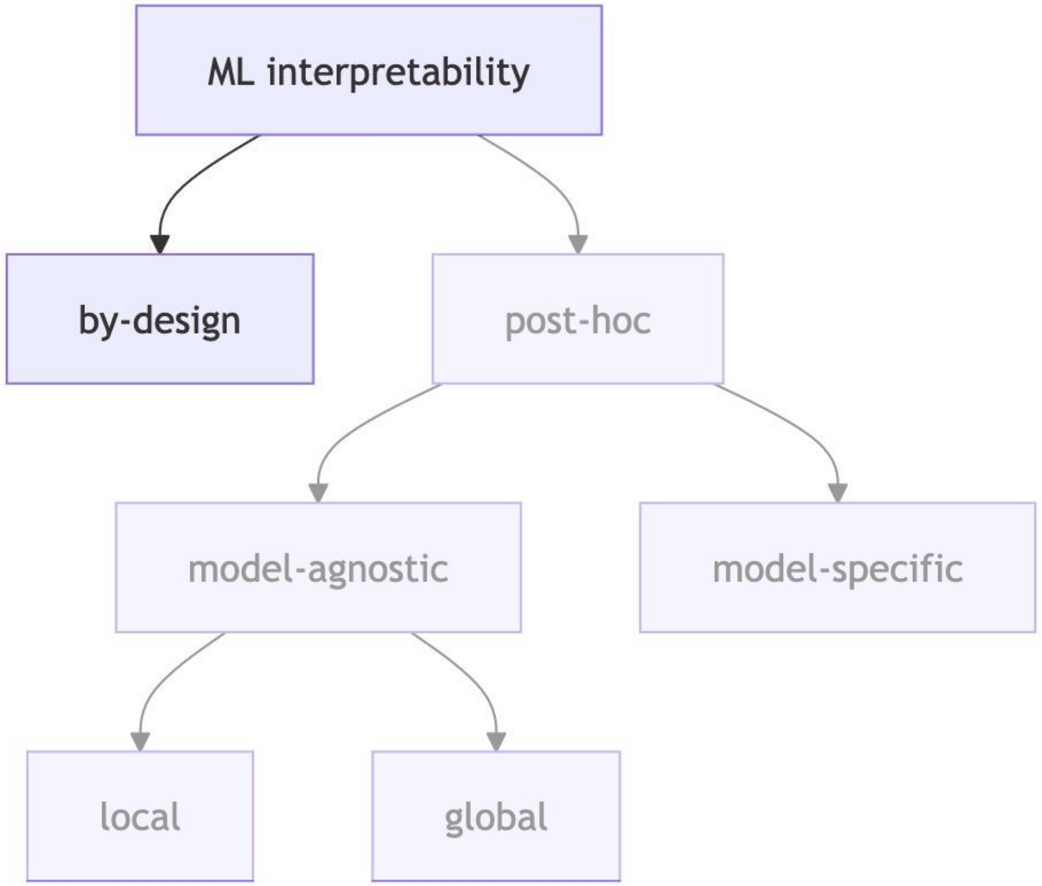
Van Noorden, R. & Perkel, J. M. AI and science: what 1,600 researchers think. *Nature* **621**, 672–675 (2023). https://doi.org/10.1038/d41586-023-02980-0

Taxonomy of interpretability methods



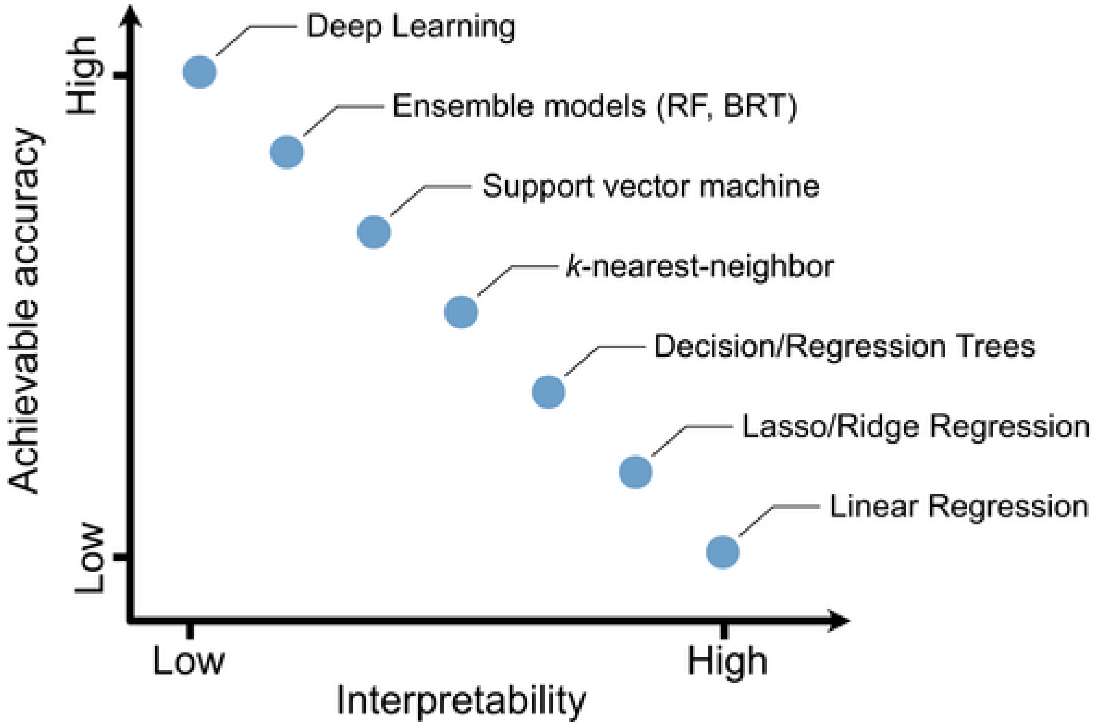
Molnar, C. *Interpretable Machine Learning: A Guide for Making Black Box Models Explainable* 3rd edn. (2025). https://christophm.github.io/interpretable-ml-book/

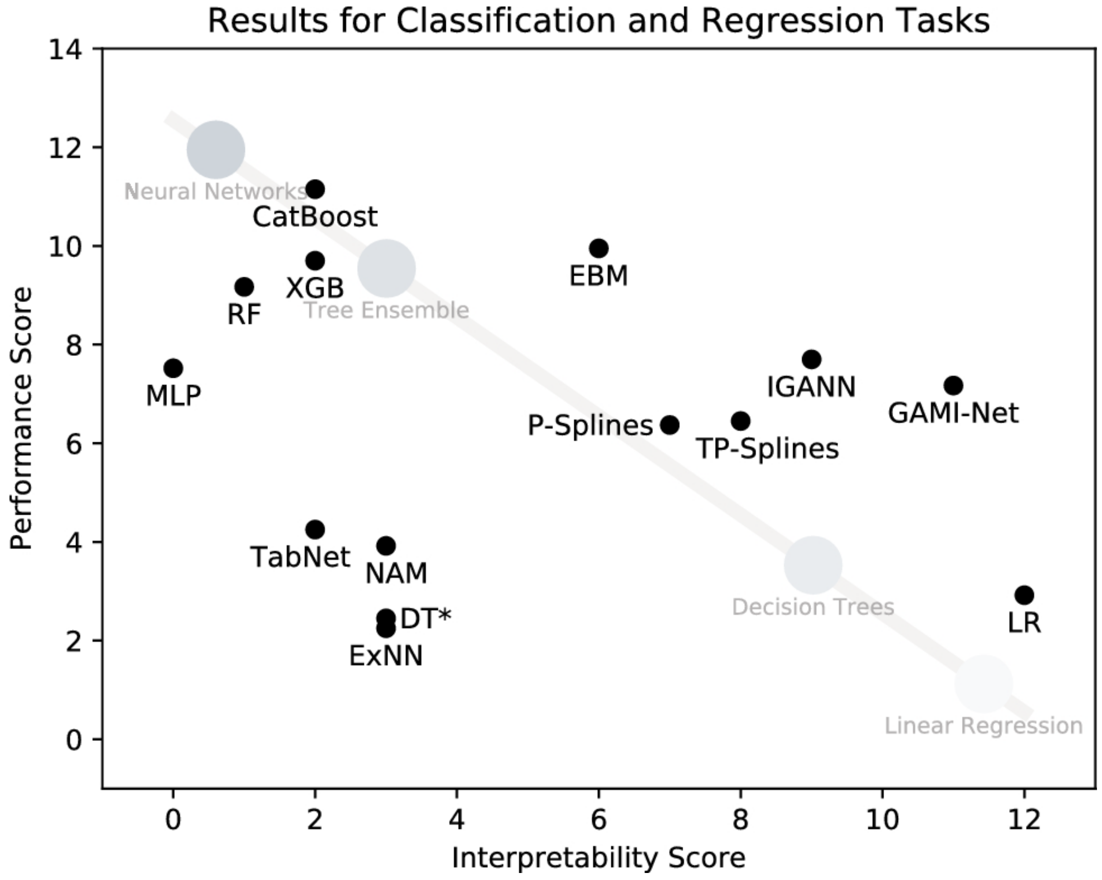
Interpretability by design



Molnar, C. *Interpretable Machine Learning: A Guide for Making Black Box Models Explainable* 3rd edn. (2025). https://christophm.github.io/interpretable-ml-book/

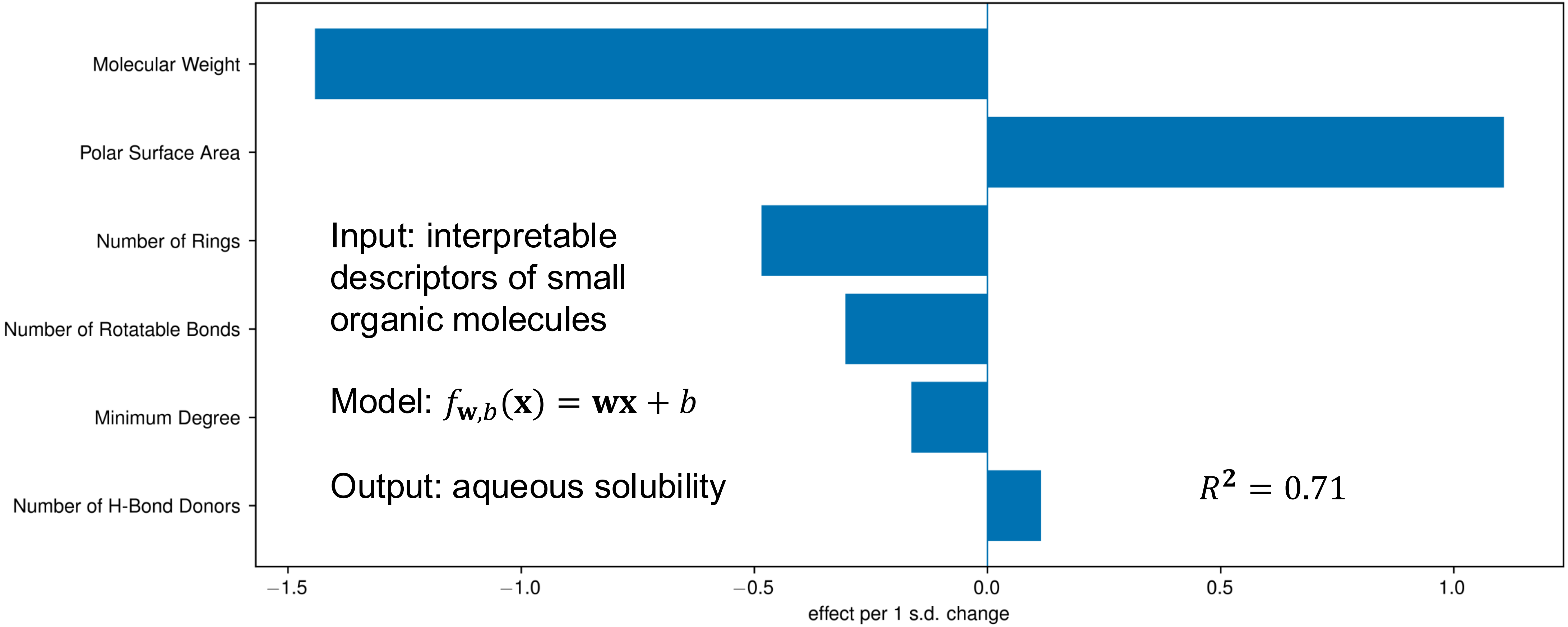
A common misconception: the performance-interpretability trade-off





Pichler, M. & Hartig, F. Machine learning and deep learning—A review for ecologists. Methods Ecol. Evol. 14, 994–1016 (2023). https://doi.org/10.1111/2041-210X.14061
Kruschel, S. *et al.* Challenging the performance-interpretability trade-off: An evaluation of interpretable machine learning models. *Bus. Inf. Syst. Eng.* **68**, 159–183 (2026). https://doi.org/10.1007/s12599-024-00922-2

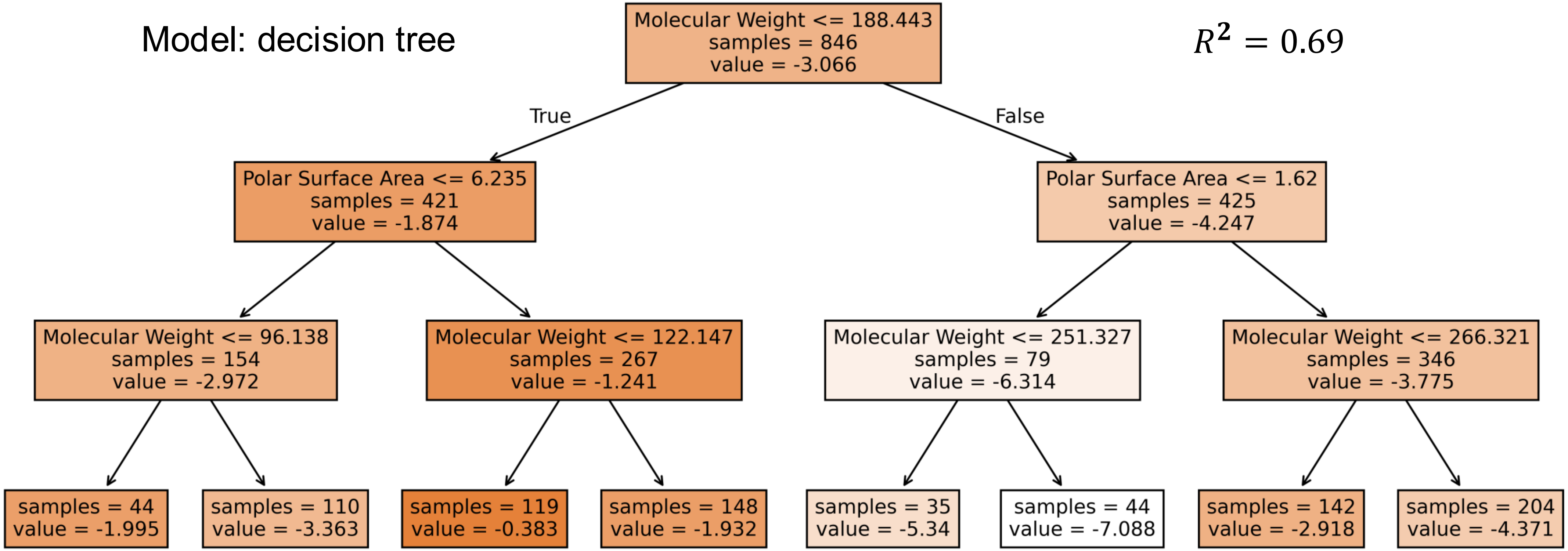
Interpretability by design: linear model coefficients



We will use the Delaney/ESOL aqueous solubility dataset for this lecture

Interpretability by design: decision trees

Model: decision tree



$R^2 = 0.69$

Interpretability by design

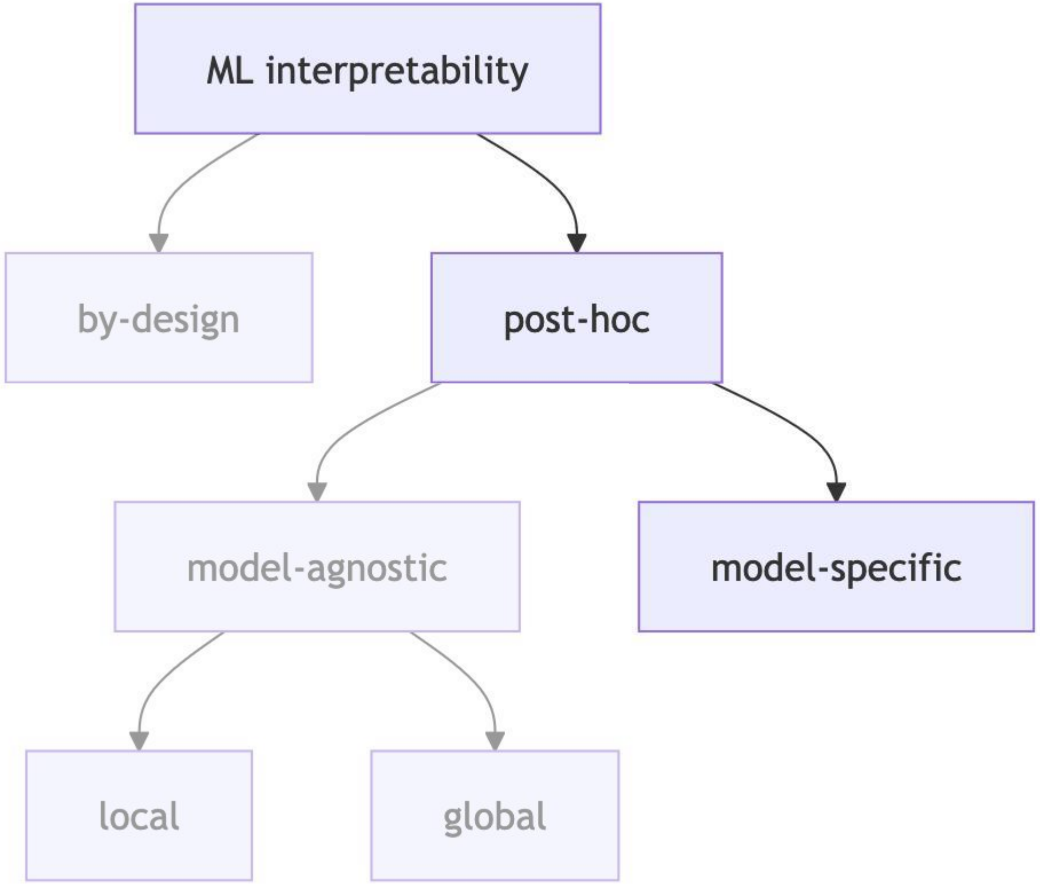
Strengths

- **Provides directly inspectable models** whose structure is understandable from the start
- **Helps reveal** simple trends, signs, rules, or constraints

Limitations

- **Does not guarantee high predictive accuracy** if the true relationship is highly nonlinear or interaction-rich
- **Can mislead when** simple coefficients or rules are treated as mechanistic truth

Model-specific post-hoc interpretability



Molnar, C. *Interpretable Machine Learning: A Guide for Making Black Box Models Explainable* 3rd edn. (2025). https://christophm.github.io/interpretable-ml-book/

Model-specific post-hoc interpretability: mean decrease in impurity (MDI) for random forests and gradient boosted trees

Definition: The impurity of a node following the squared-error criterion is the variance of the target values in that node.

$$I(t) = \frac{1}{N_t} \sum_{i \epsilon t} (y_i - \bar{y}_t)^2$$

Parent node t

where:

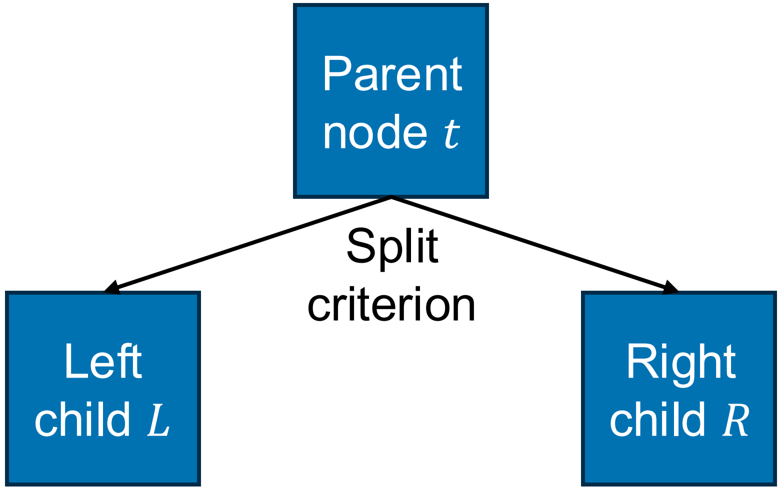
- $I(t)$: impurity/within-node variance of node t
- N_t: number of training samples in node t
- y_i: measured target value of sample i
- $\bar{y}_t$: average target value of samples in node t

Model-specific post-hoc interpretability: mean decrease in impurity (MDI) for random forests and gradient boosted trees

$$\Delta I(t) = \frac{N_t}{N}\left[I(t) - \frac{N_L}{N_t} I(L) - \frac{N_R}{N_t} I(R)\right]$$

where:

- N: total number of training samples in the tree
- N_L, N_R: number of samples in the left and right children



Model-specific post-hoc interpretability: mean decrease in impurity (MDI) for random forests and gradient boosted trees

For one tree, the importance of feature j is:

$$\text{importance}_j = \frac{\sum_{t:\text{split feature}(t)=j} \Delta I(t)}{\sum_k \sum_{t:\text{split feature}(t)=k} \Delta I(t)}$$

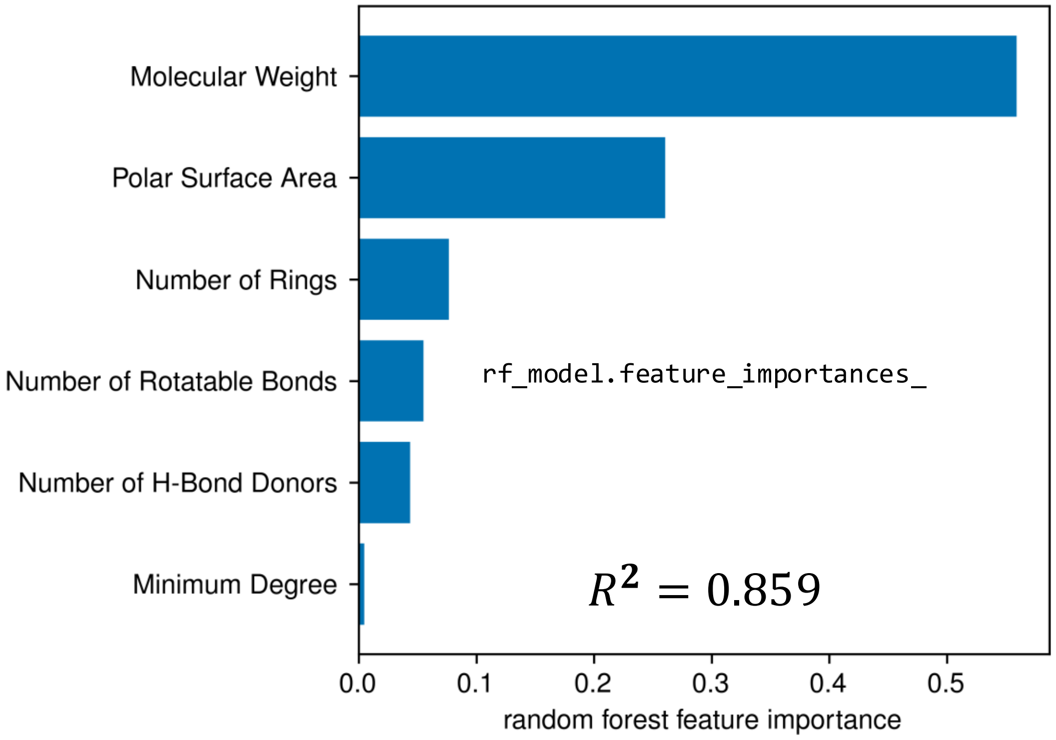
For random forests and gradient boosted tree models:

$$\text{RF/GB importance}_j = \frac{1}{T}\sum_{i=1}^{T} \text{importance}_j^{(m)}$$

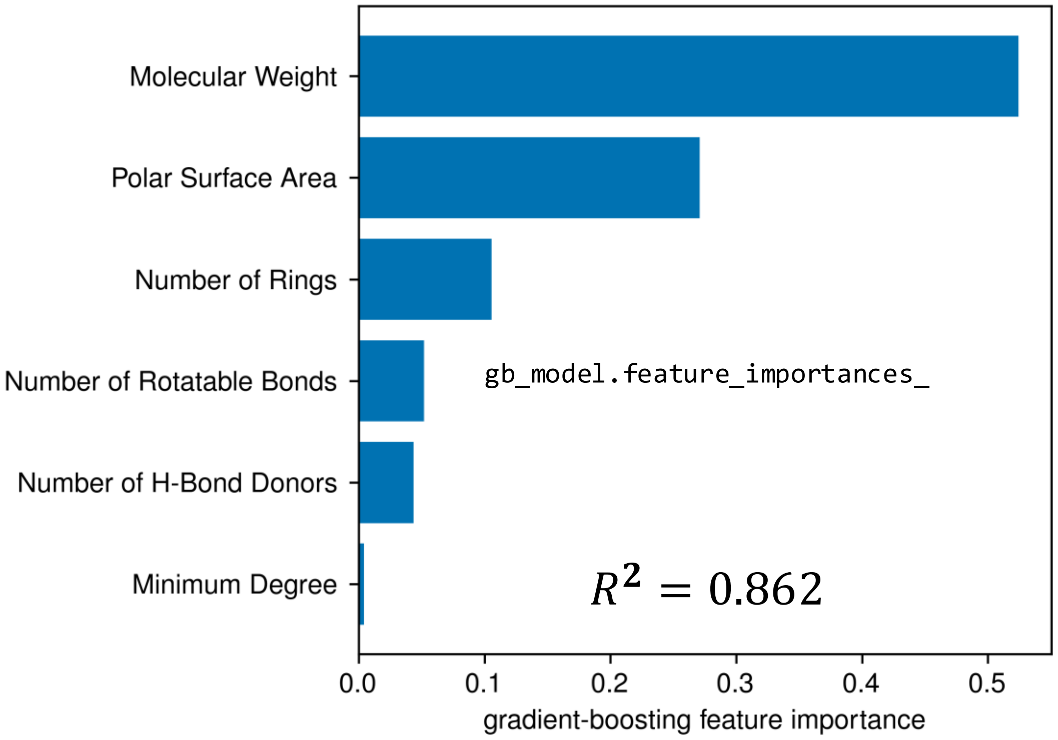
where $\text{importance}_j^{(m)}$ is the importance of feature j in tree m.

Model-specific post-hoc interpretability: mean decrease in impurity (MDI) for random forests and gradient boosted trees

Random forest



Gradient boosted tree



MDI feature importance tells us which features helped the trees reduce impurity. MDI is model-specific—it can only be applied to tree-based models.

Model-specific post-hoc interpretability: mean decrease in impurity (MDI) for random forests and gradient boosted trees

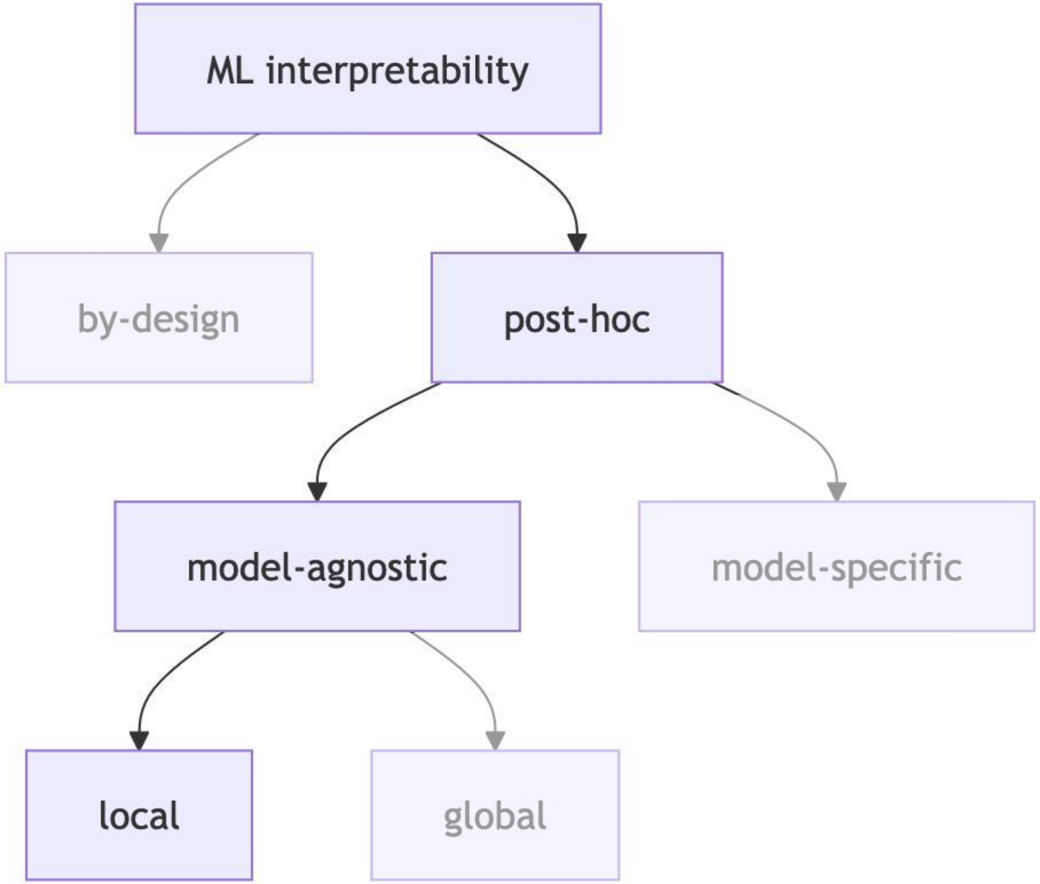
Strengths

- **Provides a fast global feature ranking** for random forests and gradient boosted trees
- **Works well when** you want a quick first look at which features helped trees make useful splits

Limitations

- **Does not show direction or local effects** because importances are unsigned and global
- **Sensitive to feature correlation** because correlated features can receive biased/split importance

Local post-hoc model-agnostic local interpretability



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Local model-agnostic post-hoc interpretability: Local Interpretable Model-agnostic Explanations (LIME)

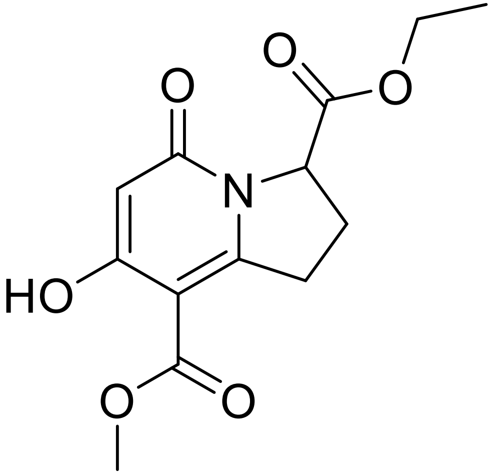
Core idea: Explain an individual prediction by fitting a simple model that mimics a complex model only near the point being explained.

Let the original model be $f(x)$ and define a simple, typically linear surrogate model $g(z)$. The goal of LIME is for g to behave like f near the data point being evaluated.

Local model-agnostic post-hoc interpretability: Local Interpretable Model-agnostic Explanations (LIME)

Example:

Step 1: Pick the molecule to explain

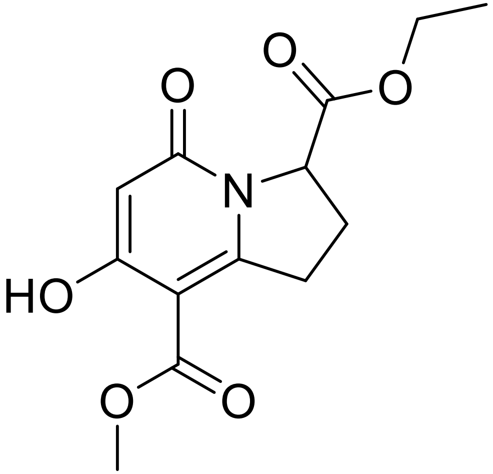


$f_{gb}(x_i) = -3.252$

Local model-agnostic post-hoc interpretability: Local Interpretable Model-agnostic Explanations (LIME)

Example:

Step 2: Generate perturbed versions of the molecule



$f_{gb}(x_i) = -3.252$

Descriptor values:

Minimum Degree	Molecular Weight	Number of H-Bond Donors	Number of Rings	Number of Rotatable Bonds	Polar Surface Area
1	261.119	1	1	2	54.86

Generate 5000 perturbed samples nearby:

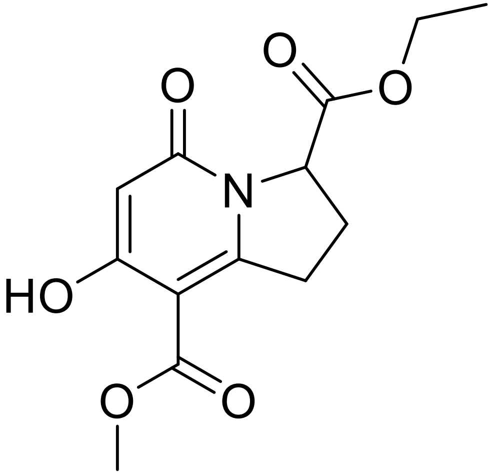
Minimum Degree	Molecular Weight	Number of H-Bond Donors	Number of Rings	Number of Rotatable Bonds	Polar Surface Area
0.997	69.567	3.301	2.000	2.564	39.467
0.911	226.866	3.222	3.064	9.390	0.000
0.918	455.888	0.000	4.111	2.352	95.249

⋮

Local model-agnostic post-hoc interpretability: Local Interpretable Model-agnostic Explanations (LIME)

Example:

Step 3: Predict the perturbed samples with f



$f_{gb}(x_i) = -3.252$

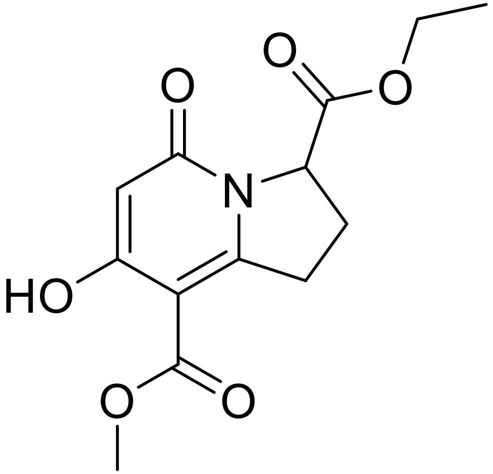
Minimum Degree	Molecular Weight	Number of H-Bond Donors	Number of Rings	Number of Rotatable Bonds	Polar Surface Area	$f(z)$
0.997	69.567	3.301	2.000	2.564	39.467	-0.060
0.911	226.866	3.222	3.064	9.390	0.000	-8.016
0.918	455.888	0.000	4.111	2.352	95.249	-5.524

⋮

Local model-agnostic post-hoc interpretability: Local Interpretable Model-agnostic Explanations (LIME)

Example:

Step 4: Compute how close each perturbed sample is to the original molecule



$f_{gb}(x_i) = -3.252$

Minimum Degree	Molecular Weight	Number of H-Bond Donors	Number of Rings	Number of Rotatable Bonds	Polar Surface Area	$f(z)$	$\pi_{x_i}(z)$
0.997	69.567	3.301	2.000	2.564	39.467	-0.060	0.641
0.911	226.866	3.222	3.064	9.390	0.000	-8.016	0.553
0.918	455.888	0.000	4.111	2.352	95.249	-5.524	0.553

⋮

A common choice of kernel is:

$$\pi_{x_i}(z) = \exp\left(-\frac{D(x_i, z)^2}{\sigma^2}\right)$$

- $\pi_{x_i}(z)$: weight assigned to perturbation sample z
- $D(x_i, z)$: Euclidean distance
- σ: kernel width

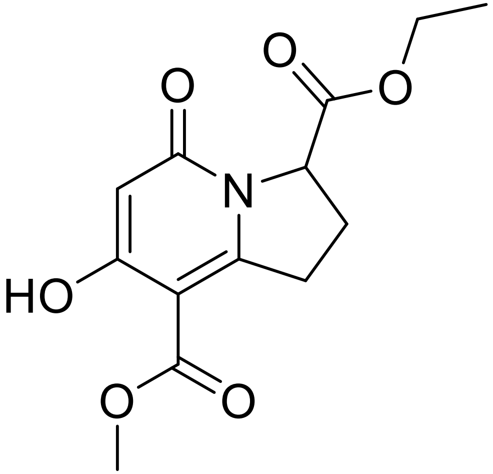
Local model-agnostic post-hoc interpretability: Local Interpretable Model-agnostic Explanations (LIME)

Example:

Step 5: Fit a simple weighted surrogate model

$$g(z) = \beta_0 + \beta_1 z_1 + \beta_1 z_2 + \cdots + \beta_p z_p$$

$$\xi(x_i) = \underset{g \in G}{\operatorname{argmin}} \sum_{k=1}^{5000} \pi_{x_i}(z_k)[f(z_k) - g(z_k)]^2$$
$$\|\beta\|_0 \leq 4$$



$$f_{gb}(x_i) = -3.252$$

$$g(x_i) = -2.809$$
$$R^2 = 0.054$$

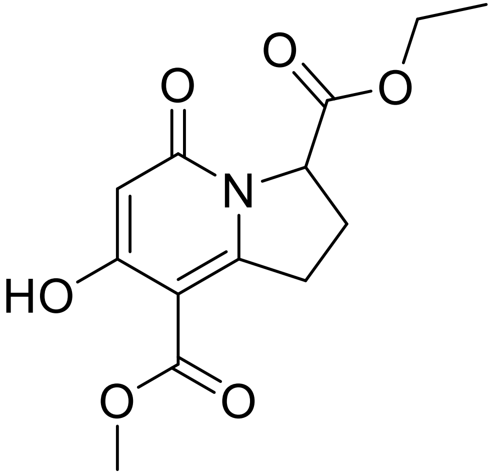
Using rule-based bins instead of feature coefficients,

$$g(z) = -3.279$$
$$+1.099 * I[26.30 < \text{Polar Surface Area} \leq 56.79]$$
$$-0.685 * I[190.35 < \text{Molecular Weight} \leq 278.06]$$
$$+0.151 * I[1.00 < \text{Number of Rotatable Bonds} \leq 3.00]$$
$$-0.067 * I[\text{Minimum Degree} \leq 1.00]$$

Local model-agnostic post-hoc interpretability: Local Interpretable Model-agnostic Explanations (LIME)

Example:

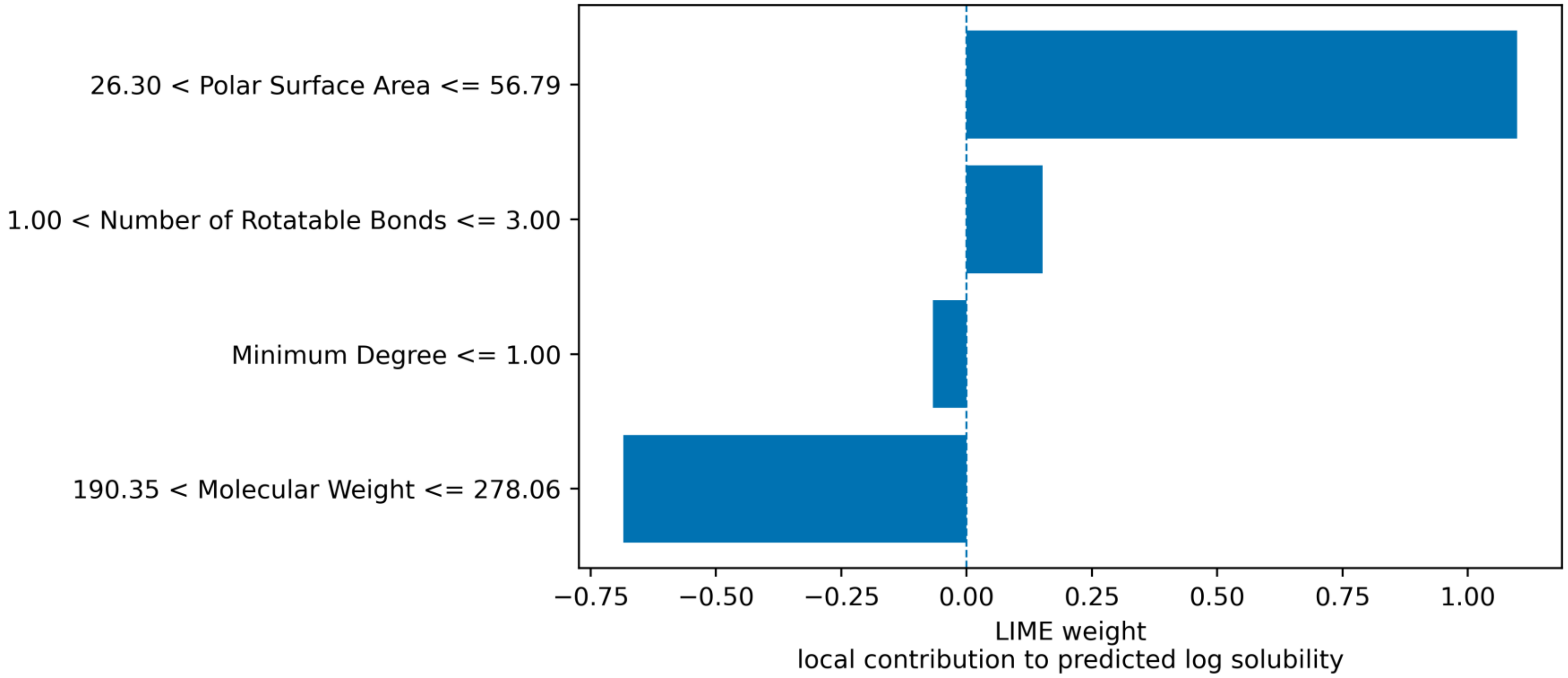
Step 5: Fit a simple weighted surrogate model



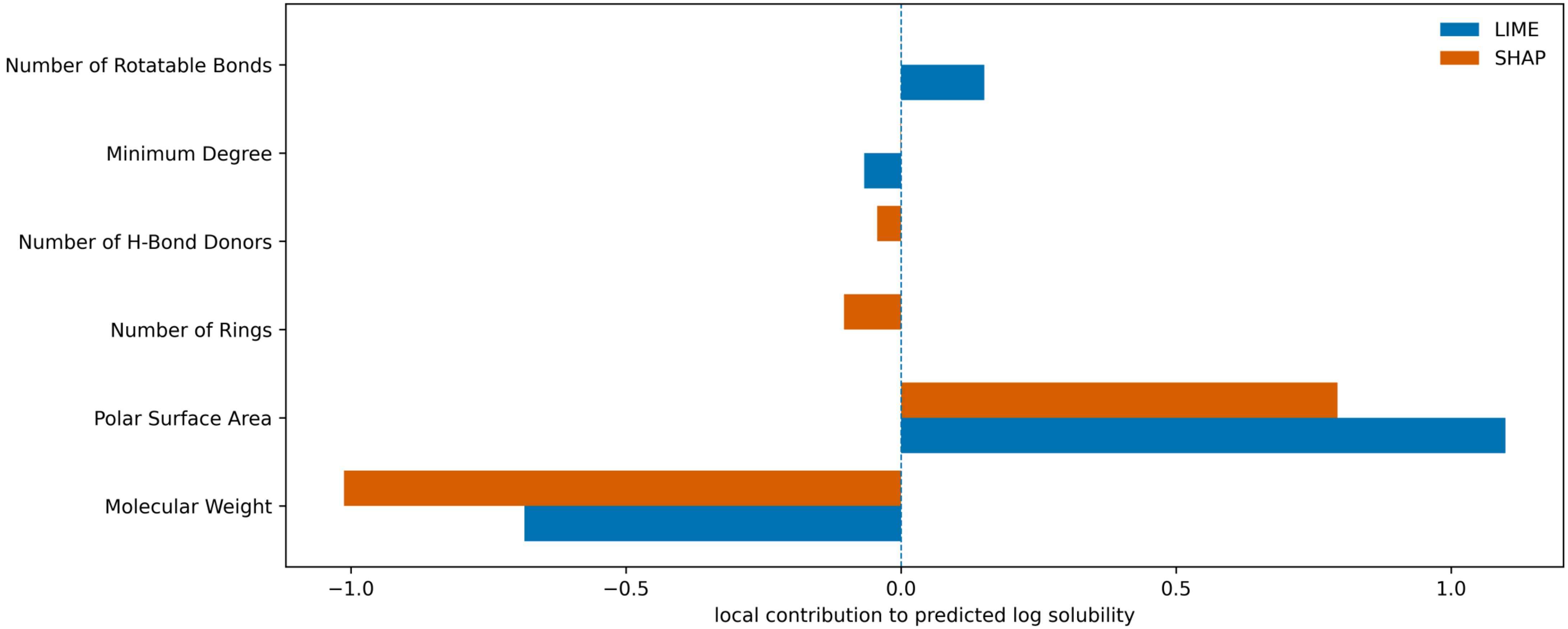
$f_{gb}(x_i) = -3.252$

$g(x_i) = -2.809$

$R^2 = 0.054$



Local model-agnostic post-hoc interpretability: Local Interpretable Model-agnostic Explanations (LIME)



Local model-agnostic post-hoc interpretability: Local Interpretable Model-agnostic Explanations (LIME)

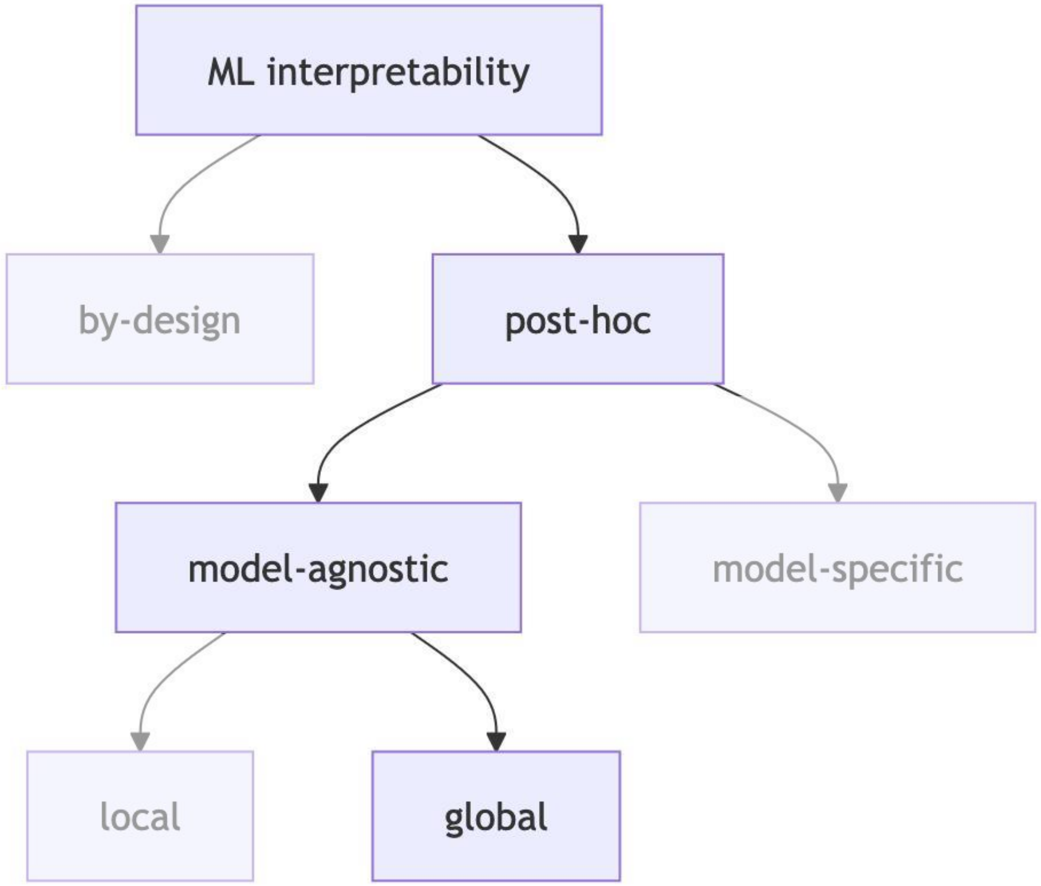
Strengths

- **Provides local, human-readable explanations** for individual predictions
- **Works well when** a simple local surrogate can approximate the black-box model near the sample being explained

Limitations

- **Does not explain the black-box model directly**; it explains a local surrogate approximation
- **Sensitive to perturbation strategy, random seed, kernel width, number of features, etc.**
- **Can mislead when** the local surrogate has poor fidelity

Global post-hoc model-agnostic local interpretability



Molnar, C. *Interpretable Machine Learning: A Guide for Making Black Box Models Explainable* 3rd edn. (2025). https://christophm.github.io/interpretable-ml-book/

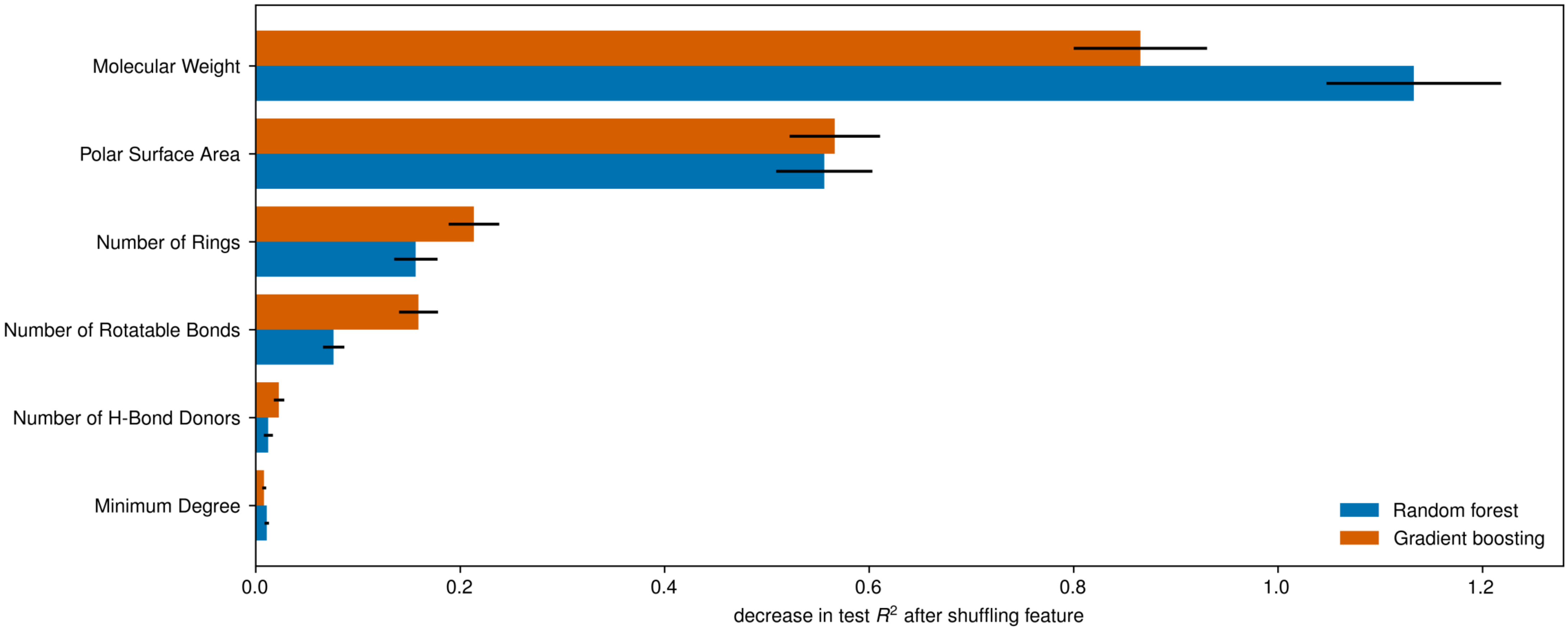
Global model-agnostic post-hoc interpretability: permutation importance

Let f be a fitted model, let X be the dataset used for inspection (usually a held-out validation or test set), and let y be the corresponding target values.

1. Compute the baseline score: $s = \mathrm{score}(f, X, y)$ (here, we will use $s = R^2(y, f(X))$).
2. Choose feature j. Randomly permute column j, leaving all other columns unchanged. Call the corrupted dataset X_j.
3. Compute the score on the corrupted dataset: $s_j = \mathrm{score}(f, X_j, y)$
4. The permutation importance of feature j is: $\mathrm{PI}_j = s - s_j$.
5. Repeat K times for each feature and average the score after shuffling:

$$\mathrm{PI}_j = s - \frac{1}{K}\sum_{k=1}^{K} s_{k,j}$$

Global model-agnostic post-hoc interpretability: permutation importance



When two features are correlated, permuting one feature may leave the model access to similar information through the correlated feature, reducing the reported importance for both.

Global model-agnostic post-hoc interpretability: permutation importance

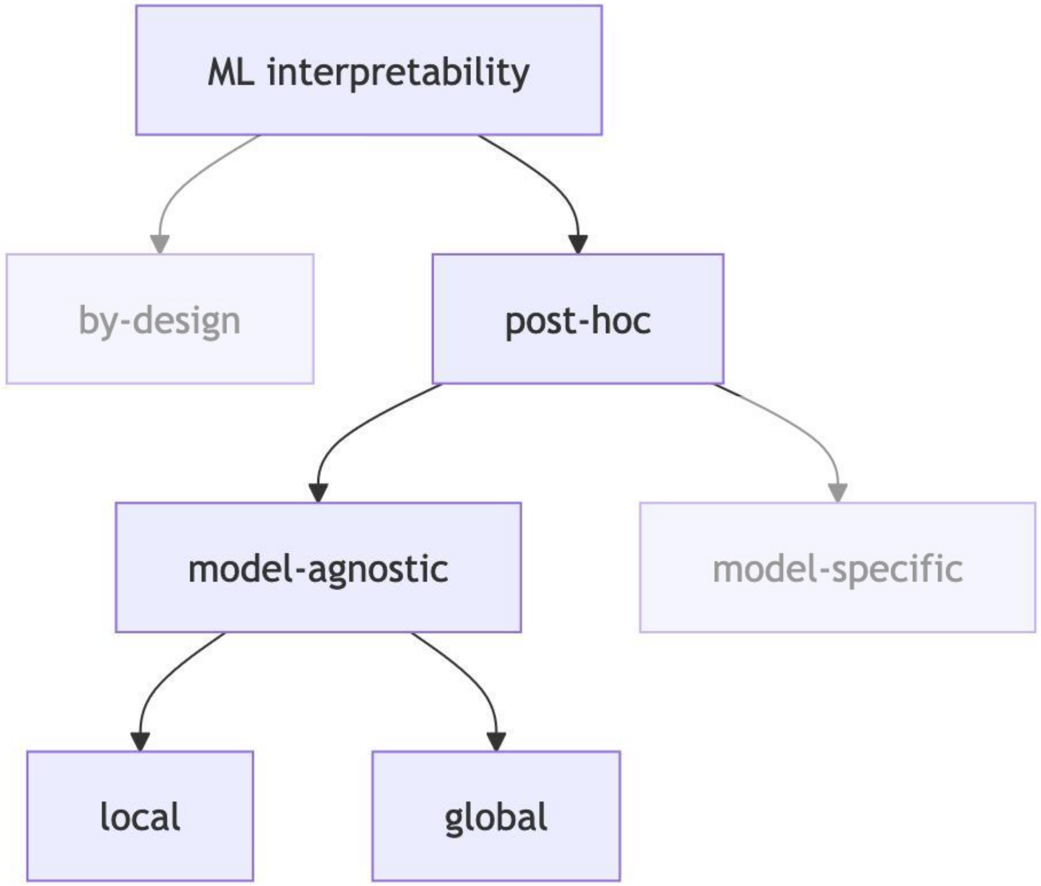
Strengths

- **Provides a model-agnostic global feature ranking** based on predictive performance
- **Helps reveal** how much performance decreases when each feature's information is destroyed

Limitations

- **Does not show direction or local effects** because it measures performance drop, not positive/negative contribution
- **Can mislead when** redundant descriptors hide each other's importance

Local & global post-hoc model-agnostic local interpretability



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Local & global model-agnostic post-hoc interpretability: individual conditional expectation (ICE) and partial dependence plots (PDP)

Let $f(x)$ be a fitted model.

For measurement i, feature j, and a chosen value z for $x_{i,j}$, the ICE value is:

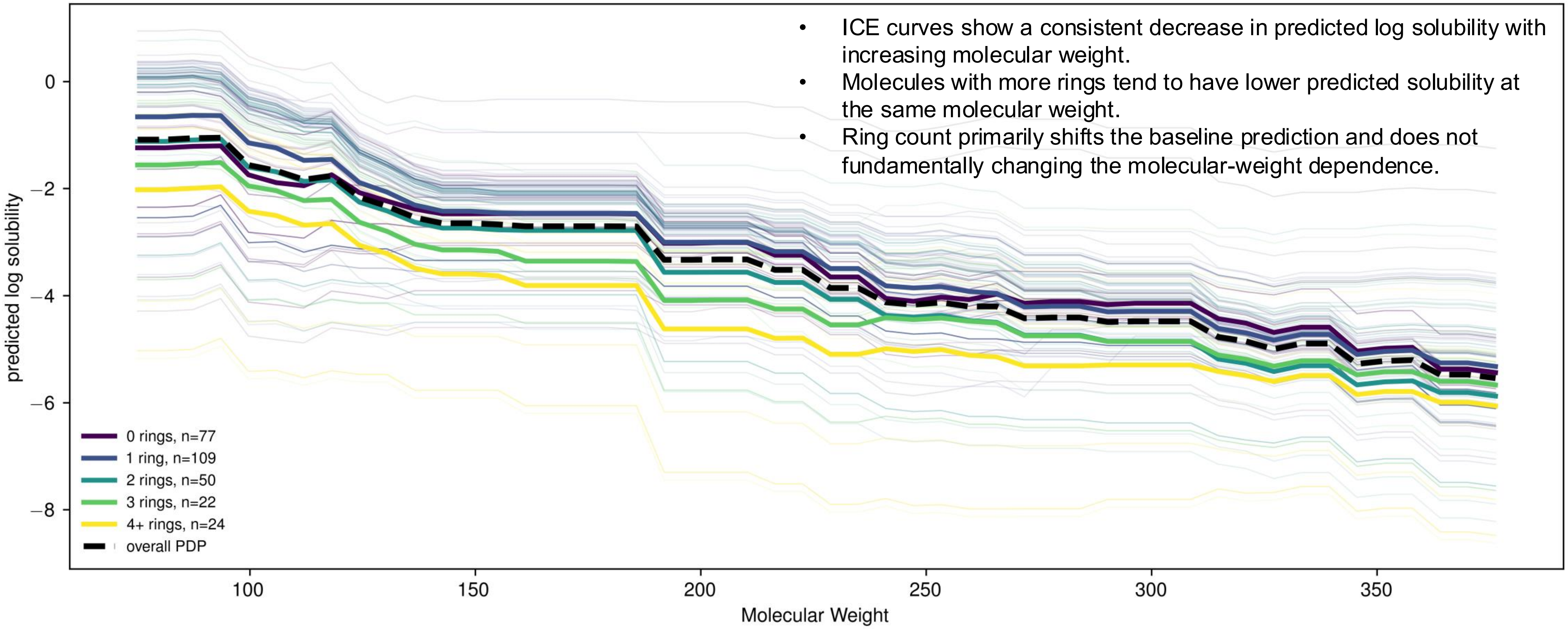
$$\mathrm{ICE}_{i,j}(z) = f(z, x_{i,-j})$$

$x_{i,-j}$: all features other than j for measurement i

A PDP averages the ICE curves over all measurements:

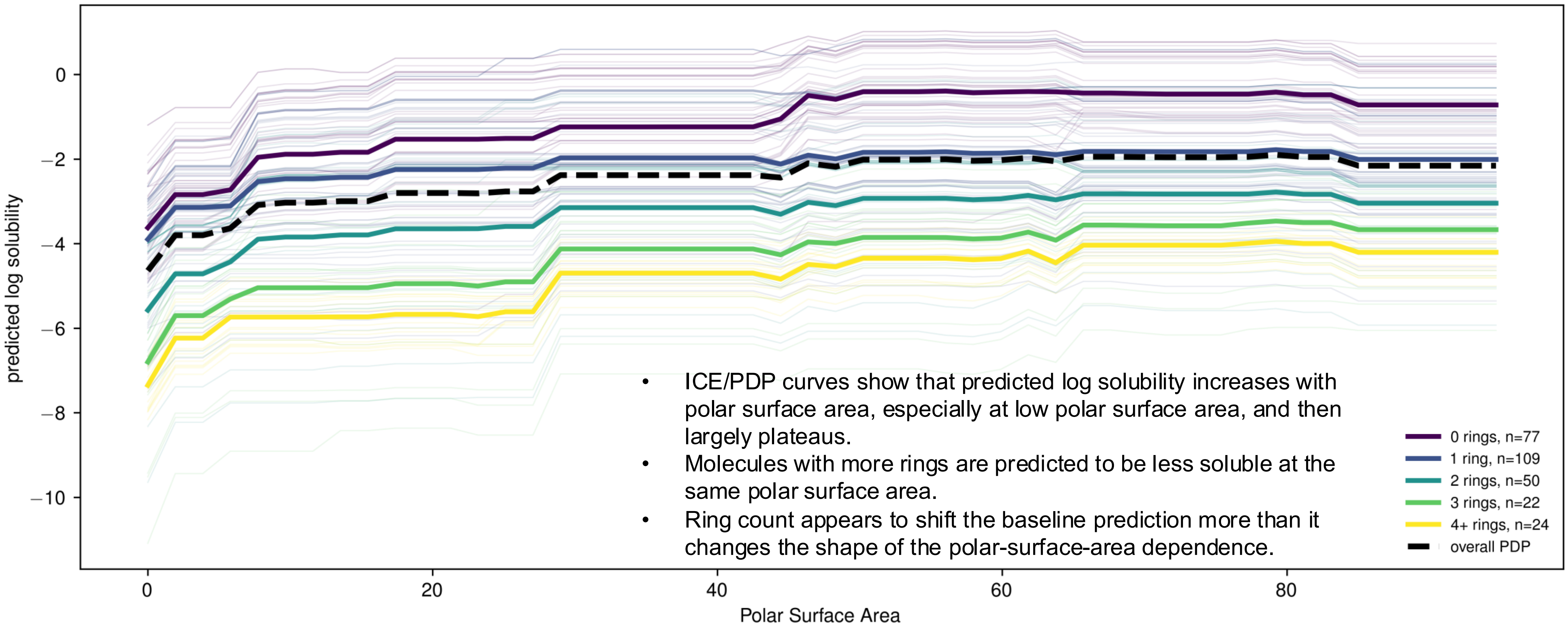
$$\mathrm{PDP}_j(z) = \frac{1}{n}\sum_{i=1}^{n} f(z, x_{i,-j})$$

Local & global model-agnostic post-hoc interpretability: individual conditional expectation (ICE) and partial dependence plots (PDP)



- ICE curves show a consistent decrease in predicted log solubility with increasing molecular weight.
- Molecules with more rings tend to have lower predicted solubility at the same molecular weight.
- Ring count primarily shifts the baseline prediction and does not fundamentally changing the molecular-weight dependence.

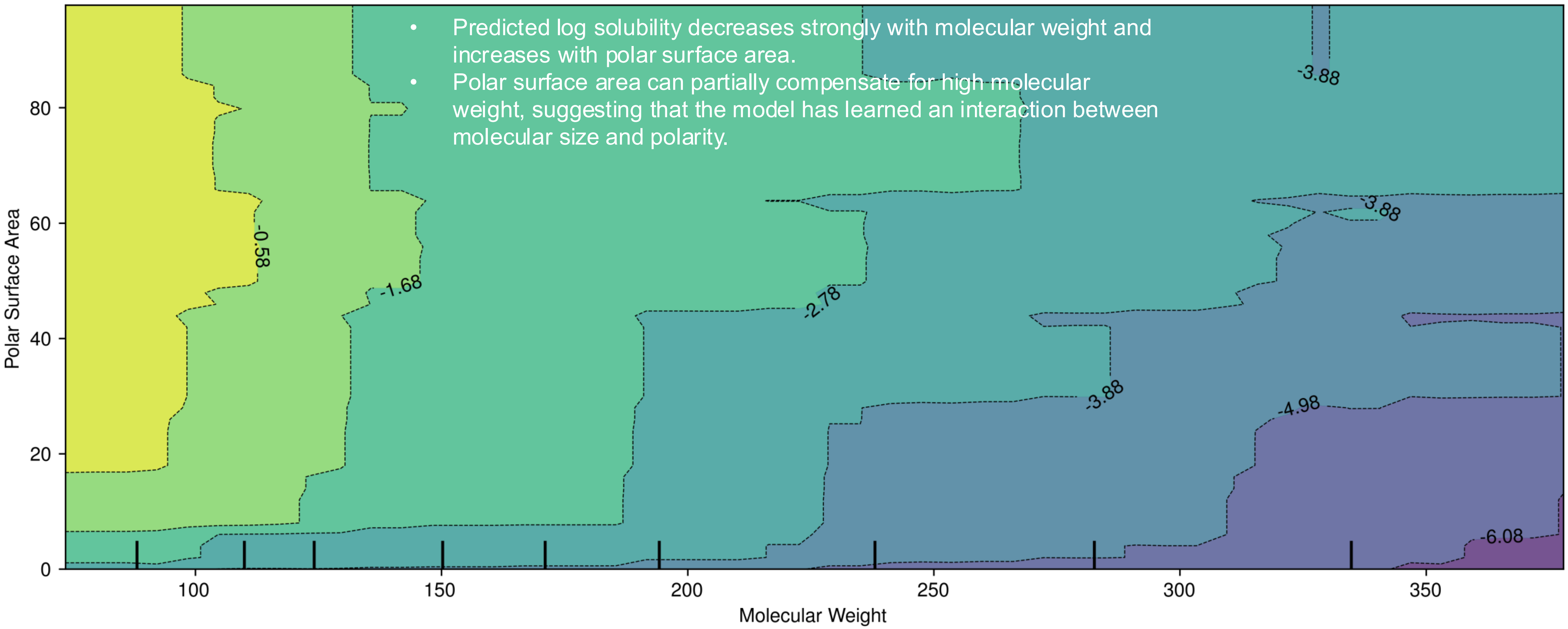
Local & global model-agnostic post-hoc interpretability: individual conditional expectation (ICE) and partial dependence plots (PDP)



- ICE/PDP curves show that predicted log solubility increases with polar surface area, especially at low polar surface area, and then largely plateaus.
- Molecules with more rings are predicted to be less soluble at the same polar surface area.
- Ring count appears to shift the baseline prediction more than it changes the shape of the polar-surface-area dependence.

Local & global model-agnostic post-hoc interpretability: individual conditional expectation (ICE) and partial dependence plots (PDP)

- Predicted log solubility decreases strongly with molecular weight and increases with polar surface area.
- Polar surface area can partially compensate for high molecular weight, suggesting that the model has learned an interaction between molecular size and polarity.



Local & global model-agnostic post-hoc interpretability: individual conditional expectation (ICE) and partial dependence plots (PDP)

Strengths

- **Works well when** you want to move from “which features matter?” to “how does the model use this feature?”
- **Helps reveal** direction, nonlinearity, thresholds, saturation, and heterogeneity across samples

Limitations

- **Do not prove causal effects** because they evaluate model behavior under artificial feature changes
- **Can mislead when** PDP averages hide heterogeneous ICE behavior

Local & global model-agnostic post-hoc interpretability: SHapley Additive exPlanations (SHAP)

Core idea: every prediction is decomposed as

$$f(x_i) = \varphi_0 + \varphi_{i,1} + \varphi_{i,2} + \cdots + \varphi_{i,p}$$

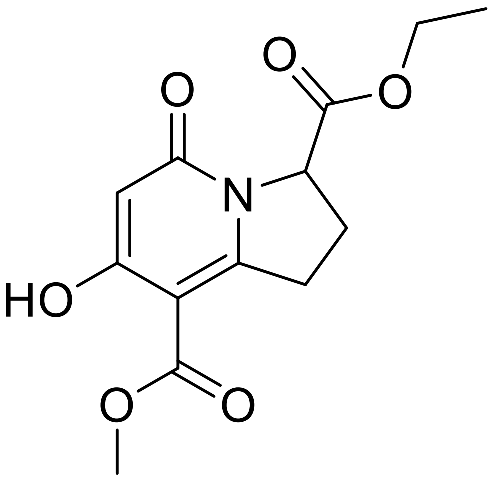
where

- $f(x_i)$: model prediction for sample i
- φ_0: baseline prediction
- $\varphi_{i,j}$: SHAP value for feature j in sample i
- p: number of features

Local & global model-agnostic post-hoc interpretability: SHapley Additive exPlanations (SHAP)

Example:

$f_{gb}(x_i) = -3.252$



$\varphi_0 = \mathbb{E}[f(X)] = -2.885$

$$\therefore \sum \varphi_{i,j} = \varphi_{i,\mathrm{mw}} + \varphi_{i,\mathrm{psa}} + \varphi_{i,\#\mathrm{r}} + \varphi_{i,\#\mathrm{hbd}} + \varphi_{i,\mathrm{md}} + \varphi_{i,\#\mathrm{rb}}$$

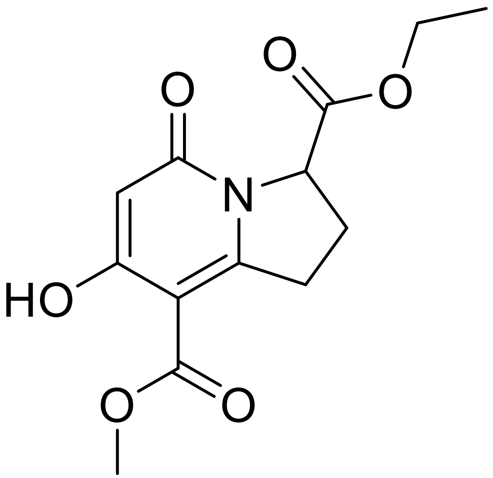
$$= -3.252 - (-2.885) = -0.367$$

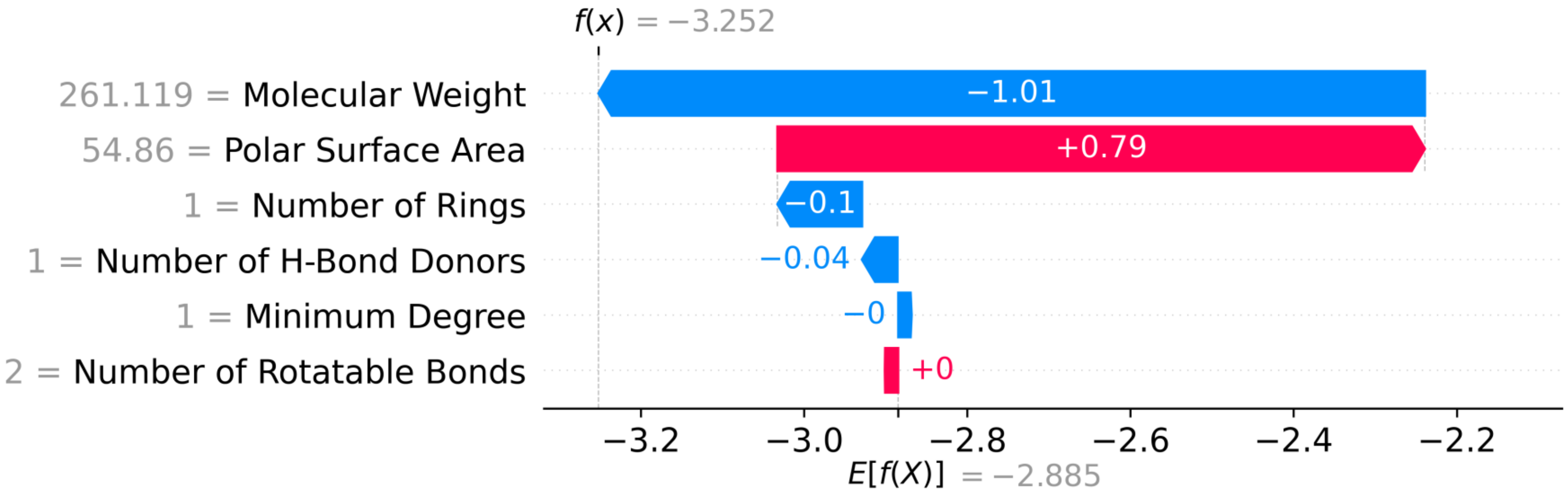
How much did molecular weight, polar surface area, ring count, and the other descriptors push the predicted log solubility above or below a reference prediction?

Local & global model-agnostic post-hoc interpretability: SHapley Additive exPlanations (SHAP)

Example:

$f_{gb}(x_i) = -3.252$



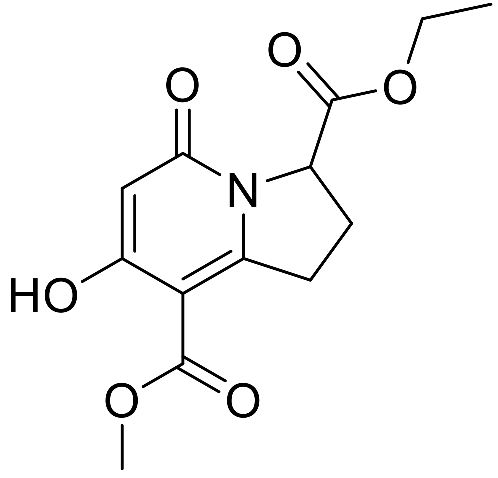


How much did molecular weight, polar surface area, ring count, and the other descriptors push the predicted log solubility above or below a reference prediction?

Local & global model-agnostic post-hoc interpretability: SHapley Additive exPlanations (SHAP)

Example:

Step 1: Pick the molecule to explain



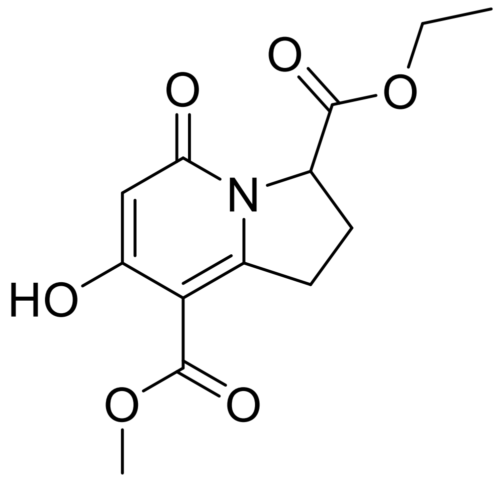
$f_{gb}(x_i) = -3.252$

Local & global model-agnostic post-hoc interpretability: SHapley Additive exPlanations (SHAP)

Example:

Step 2: Choose a background dataset X

$X =$ `X_train`



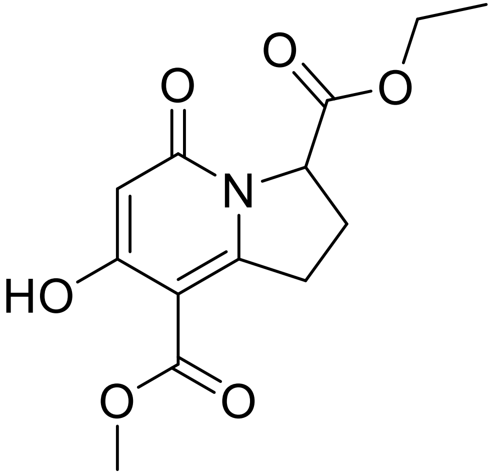
$f_{gb}(x_i) = -3.252$

Local & global model-agnostic post-hoc interpretability: SHapley Additive exPlanations (SHAP)

Example:

Step 3: Compute the baseline prediction

$X =$ `X_train`



$\varphi_0 = \mathbb{E}[f(X)] = -2.885$

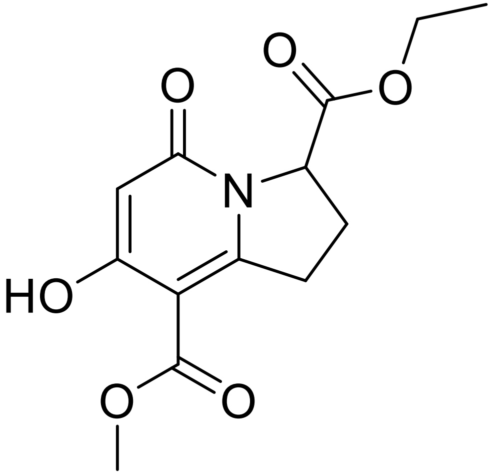
$f_{gb}(x_i) = -3.252$

Local & global model-agnostic post-hoc interpretability: SHapley Additive exPlanations (SHAP)

Example:

Step 4: Consider all feature coalitions without molecular weight

$X =$ `X_train`



$f_{gb}(x_i) = -3.252$

$\varphi_0 = \mathbb{E}[f(X)] = -2.885$

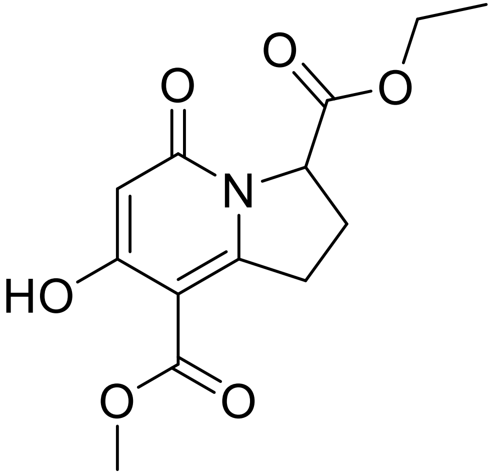
{}
{Polar Surface Area}
{Number of Rings}
{Polar Surface Area, Number of Rings}
{Number of Rings, Number of Rotatable Bonds}
...
{Polar Surface Area, Number of Rings, Rotatable Bonds, H-Bond Donors, Minimum Degree}

$2^{6-1} = 32$ total coalitions

Local & global model-agnostic post-hoc interpretability: SHapley Additive exPlanations (SHAP)

Example:

Step 5: Measure what happens when molecular weight is added



$f_{gb}(x_i) = -3.252$

$X =$ `X_train`

$\varphi_0 = \mathbb{E}[f(X)] = -2.885$

{}
{Polar Surface Area}
{Number of Rings}
{Polar Surface Area, Number of Rings}
{Number of Rings, Number of Rotatable Bonds}
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{Polar Surface Area, Number of Rings, Rotatable Bonds, H-Bond Donors, Minimum Degree}

$2^{6-1} = 32$ total coalitions

Compute $v(S \cup \{\mathrm{MW}\}) - v(S)$ for each subset S
E.g.,

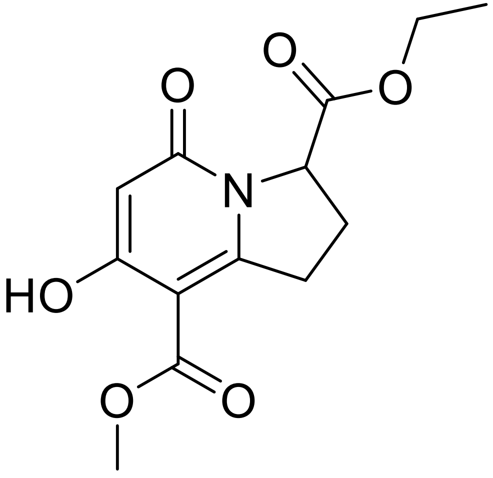
$$v(\{\mathrm{Rings}, \mathrm{MW}\}) - v(\{\mathrm{Rings}\}) = -1.02$$

$$v(\{\mathrm{PSA}, \mathrm{Rings}, \mathrm{MW}\}) - v(\{\mathrm{PSA}, \mathrm{Rings}\}) = -0.95$$

Local & global model-agnostic post-hoc interpretability: SHapley Additive exPlanations (SHAP)

Example:

Step 6: Average all marginal contributions with Shapley weights



$f_{gb}(x_i) = -3.252$

$X =$ `X_train`

$\varphi_0 = \mathbb{E}[f(X)] = -2.885$

{}
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{Number of Rings}
{Polar Surface Area, Number of Rings}
{Number of Rings, Number of Rotatable Bonds}
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$2^{6-1} = 32$ total coalitions

Compute $v(S \cup \{\mathrm{MW}\}) - v(S)$ for each subset S

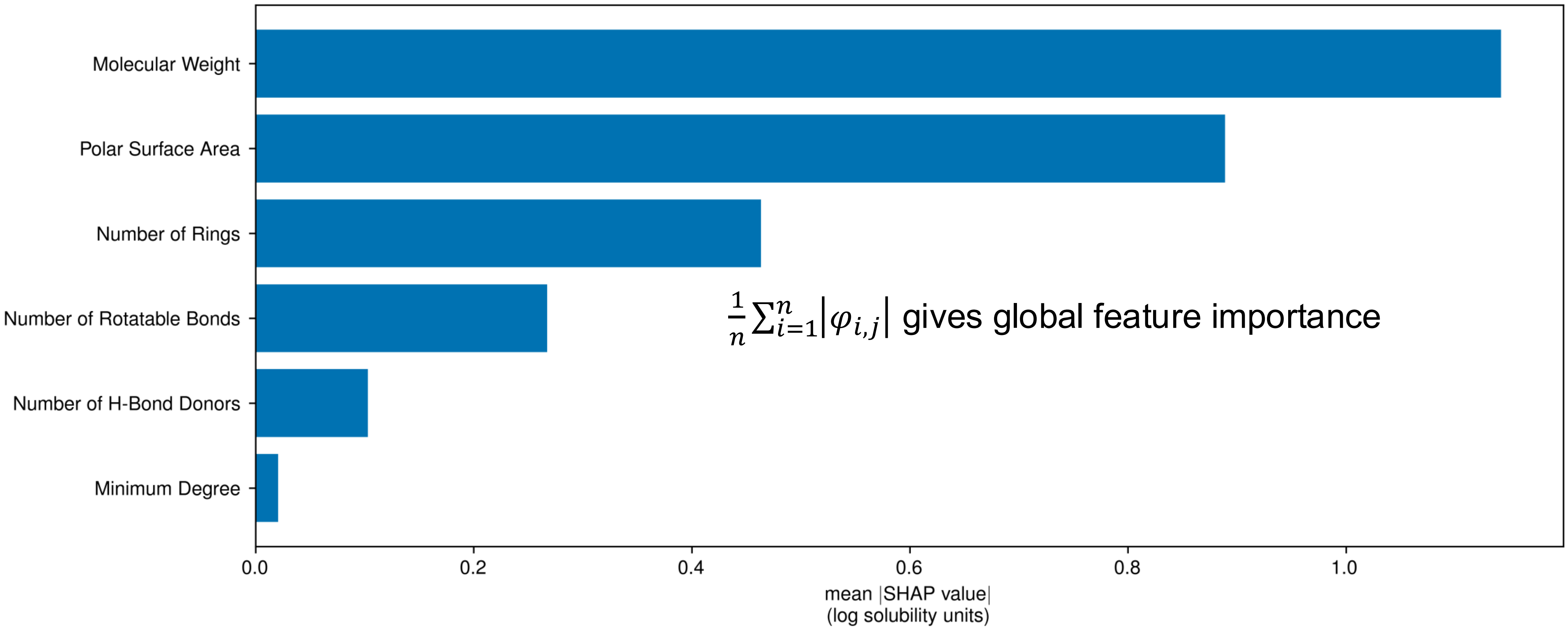
E.g.,

$$v(\{\mathrm{Rings}, \mathrm{MW}\}) - v(\{\mathrm{Rings}\}) = -1.02$$

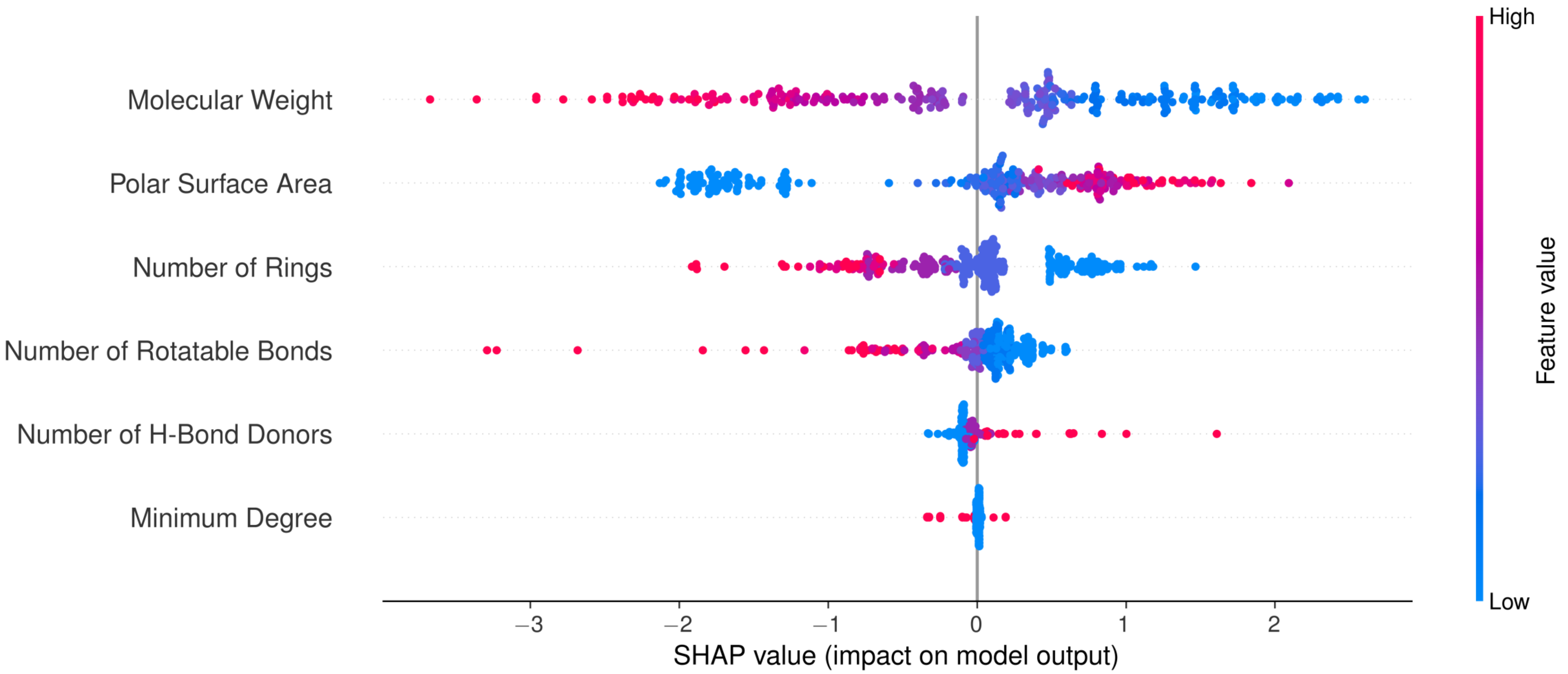
$$v(\{\mathrm{PSA}, \mathrm{Rings}, \mathrm{MW}\}) - v(\{\mathrm{PSA}, \mathrm{Rings}\}) = -0.95$$

$$\varphi_{\mathrm{MW}} = \sum_{S \subseteq M \setminus \{\mathrm{MW}\}} \frac{|S|!\,(p - |S| - 1)!}{p!} [v(S \cup \{\mathrm{MW}\}) - v(S)] = -1.01$$

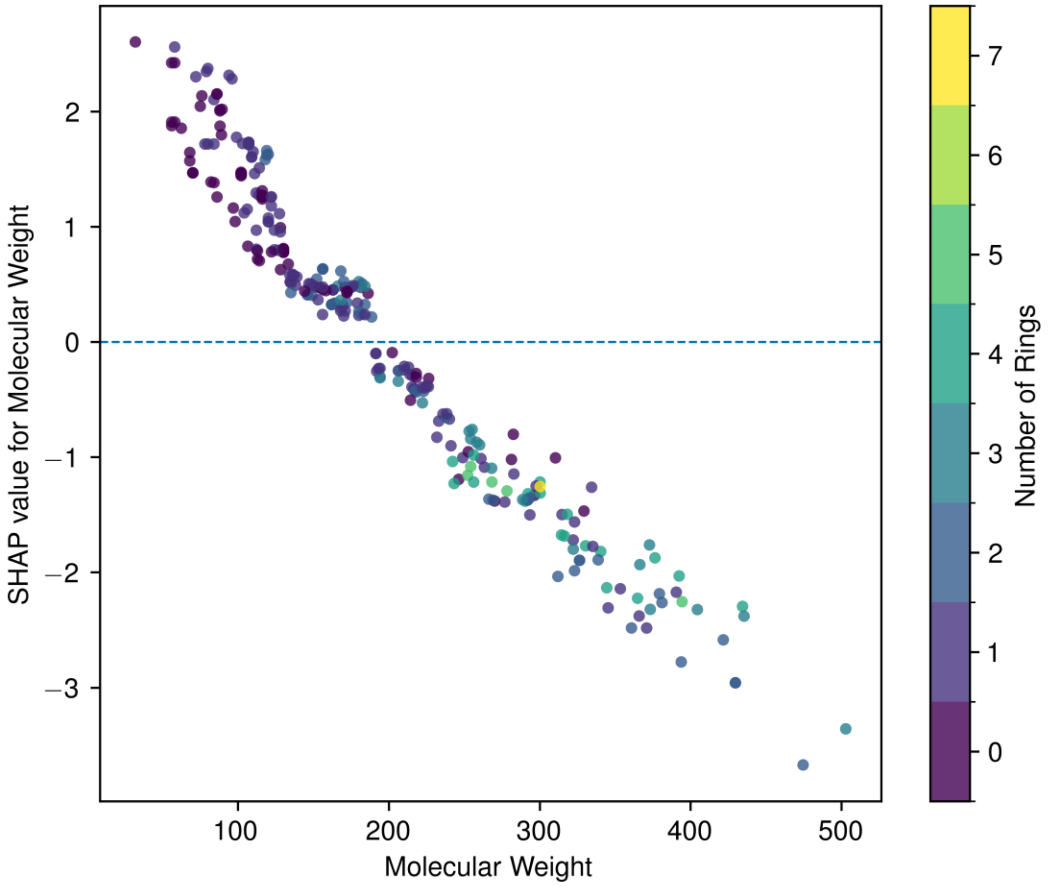
Local & global model-agnostic post-hoc interpretability: SHapley Additive exPlanations (SHAP)

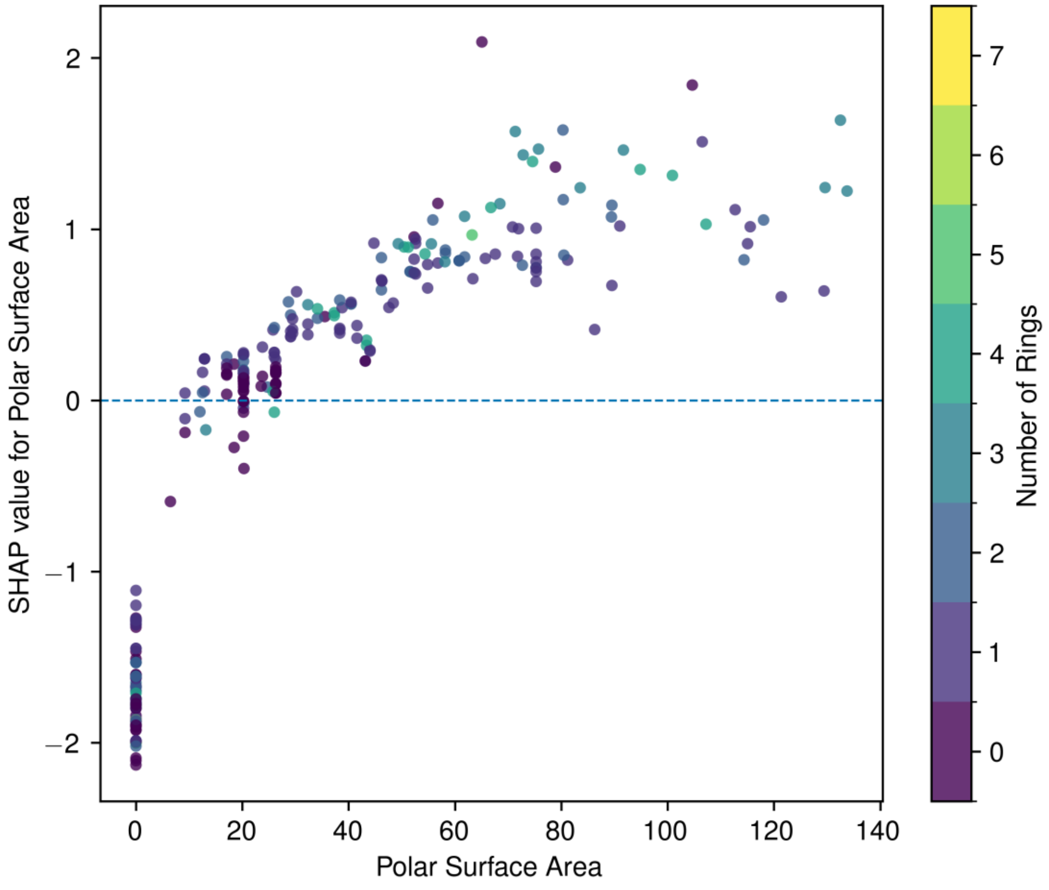


Local & global model-agnostic post-hoc interpretability: SHapley Additive exPlanations (SHAP)



Local & global model-agnostic post-hoc interpretability: SHapley Additive exPlanations (SHAP)





Local & global model-agnostic post-hoc interpretability: SHapley Additive exPlanations (SHAP)

Strengths

- **Provides local additive feature attributions** for individual predictions
- **Works well when** you want both local explanations and global summaries from aggregated SHAP values
- **Helps reveal** direction, magnitude, and distribution of feature contributions across samples

Limitations

- **Does not prove mechanism or causality**; it explains the model's attribution of a prediction
- **Sensitive to correlated feature and background/reference distribution**

Closing thoughts

Choosing the right model

Question	Method
Which features matter globally?	MDI, permutation importance, mean |SHAP|
How does a feature affect predictions?	ICE/PDP
Why this specific prediction?	LIME, SHAP
Does the model learn chemistry or shortcuts?	Compare methods + domain knowledge

A practical workflow

1. Start with an interpretable baseline
2. Compare against flexible models
3. Use global explanation methods to identify model reliance
4. Use local explanations to inspect examples and outliers
5. Check whether explanations make chemical sense

Common failure modes

- correlated descriptors split or hide importance
- local explanations are treated as global rules
- feature attributions are interpreted as mechanisms
- visually convincing plots create false confidence

Additional resources

Accumulated local effects (ALE)

Feature-effect plots that reduce extrapolation issues when features are correlated.

Apley & Zhu, JRSS B 2020

Explainable Boosting Machines

Interpretable GAM-like models with learned nonlinear feature functions and interactions.

interpret.ml/docs/ebm.html

GNNExplainer

Subgraph and node-feature explanations for graph neural networks.

arxiv.org/abs/1903.03894

Use interpretable models when possible

A cautionary perspective on post-hoc explanations in high-stakes settings.

Rudin, Nat. Mach. Intell. 2019

Counterfactual explanations

Ask: "What minimal molecular/property change would flip the prediction?"

interpret.ml/DiCE

Integrated gradients / Captum

Attribution methods for neural networks; useful for differentiable molecular models.

captum.ai

Sanity checks for explanations

Test whether saliency/explanation maps actually depend on the model and data.

Adebayo et al., NeurIPS 2018

Benchmark explanation quality

Libraries such as Quantus help compare robustness and faithfulness of explanations.

quantus.readthedocs.io